

Privacy-Preserving Vertical Federated Learning

for Joint Linear Regression when the Organization Holds the Response

Working Draft — revised protocol

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Abstract

We study two-party vertical federated ordinary least squares (OLS) between a researcher R and an organization O (e.g. a national statistical agency) in the regime where the response $\mathbf{y}$ is held by O . Records are linked by labeled private set intersection, the O -dependent blocks of the joint Gram matrix are aggregated under homomorphic encryption, and differential privacy (DP) noise is injected in the encrypted domain; R 's own Gram block is computed locally and never transmitted. Relative to earlier drafts, the protocol is revised to close gaps that invalidate the stated guarantees against an honest-but-curious R : O 's replies are sanitized (fresh re-encryption and noise flooding), because a plaintext-scalar noise addition leaves the ciphertext's randomness a deterministic function of O 's data; the noise is calibrated to the replace-one sensitivity that the adjacency relation actually requires; calibration is split between the parties so that only committed public constants are exchanged; O keeps a zero-concentrated DP ledger across runs; and the ridge parameter is fixed from public inputs. We then characterize the finite-noise bias of the private estimator as a conditional expectation, give a correction that costs no privacy budget, and state the remaining proof obligations explicitly. Empirically, the cryptographic layer reproduces plaintext OLS to about 10^{-10} . On real data the private estimate comes within 10% of non-private test error at 1.4k–11k matched records for $\epsilon \in \{1, 2\}$; against centralized AdaSSP on 29 UCI datasets it is ahead at small ϵ and needs 1.4–2.1 times the records at moderate ϵ , and under the same guarantee it beats AdaSSP's adaptive ridge on 27 of 29 datasets. Coefficient-level accuracy needs roughly ten times more data than prediction, and the bias correction changes test error by under 4%.

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1 Introduction

Organizations such as Statistics Canada (StatCan) hold administrative and survey datasets that span the full Canadian population. University researchers, by contrast, often hold smaller cohort datasets enriched with domain-specific variables—clinical measurements, survey responses, or behavioural signals—that StatCan does not collect. Neither dataset alone is sufficient to answer certain research questions: a policy researcher studying labour-market outcomes might hold a cohort’s health and education records, while StatCan holds their tax filing history, employment insurance claims, and income trajectories. The scientific value of joining these datasets is evident; the barrier to doing so is equally clear. Privacy law, institutional policy, and the practical realities of data governance make direct data sharing infeasible.

The Research Data Centre (RDC) model partially addresses this by letting researchers work with anonymized StatCan data inside a controlled environment. However, it does not permit a researcher to bring their own dataset and perform a record linkage against StatCan records in order to jointly train a statistical model—a process that typically requires six to eight weeks of vetting, human review, and institutional approvals. This overhead is a meaningful bottleneck on the pace and scope of Canadian policy research.

This document proposes a cryptographic protocol for *vertical federated learning* (VFL) between a researcher and StatCan. The term “vertical” refers to the structure of the data: both parties hold overlapping sets of individuals (rows) but non-overlapping sets of variables (columns). The goal is to jointly train a regression model over the combined column space without either party ever observing the other’s raw data. The two datasets are linked by a common identifier (e.g. a hashed Social Insurance Number), but the linkage itself must remain private—StatCan should not learn which of its records belong to the researcher’s cohort.

Scope. We treat the case in which **StatCan holds the response variable y** (e.g. tax-reported income, employment status, benefit receipt), and the researcher wishes to learn whether their cohort-specific covariates predict that outcome, using StatCan’s covariates as controls. The protocol combines *homomorphic encryption* (HE) and *private set intersection* (PSI) for input privacy with *differential privacy* (DP) for output privacy.

Contributions.

1. A corrected data flow in which R’s Gram block $\mathbf{A}$ is computed locally and never transmitted or noised (Section 4).
2. A revised protocol that is sound against an honest-but-curious R who holds the secret key: sanitization of every ciphertext $\mathbf{O}$ returns (Section 6), replace-one

sensitivity and a privacy ledger (Section 5), per-party calibration, and a PSI whose labels and metadata are data-independent (Appendix B). Pseudocode is given in Algorithms 1–5.

3. The leading-order bias of the private OLS and ridge estimators, stated as conditional expectations with explicit validity conditions, and a bias correction that is pure post-processing (Section 7).
4. An empirical evaluation: exactness of the cryptographic layer, coefficient accuracy against intersection size, the practical value of the correction, a benchmark against AdaSSP on 29 UCI datasets, and an intersection-size sweep on real data (Section 9).

2 Problem Setup and Notation

Let $\mathbf{R}$ denote the researcher and $\mathbf{O}$ denote the organization (StatCan). Table 1 summarises the notation used throughout the document.

The target estimator is the OLS solution on the matched, column-joined data:

$$\hat{\beta} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{y}^I, \quad (1)$$

where the Gram matrix and moment vector decompose as

$$\mathbf{G} := \mathbf{Z}^\top \mathbf{Z} = \begin{bmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{B} \end{bmatrix}, \quad \mathbf{c} := \mathbf{Z}^\top \mathbf{y}^I = \begin{bmatrix} \mathbf{c}_R \\ \mathbf{c}_O \end{bmatrix}, \quad (2)$$

with $\mathbf{A} = \mathbf{X}_R^{I\top} \mathbf{X}_R^I$, $\mathbf{B} = \mathbf{X}_O^{I\top} \mathbf{X}_O^I$, $\mathbf{C} = \mathbf{X}_R^{I\top} \mathbf{X}_O^I$, $\mathbf{c}_R = \mathbf{X}_R^{I\top} \mathbf{y}^I$, and $\mathbf{c}_O = \mathbf{X}_O^{I\top} \mathbf{y}^I$. We write Π_R and Π_O for the orthogonal projectors onto the $\mathbf{R}$ and $\mathbf{O}$ coordinate blocks, so $\Pi_R + \Pi_O = \mathbf{I}_p$.

3 Input and Output Privacy: Definitions and Threat Model

Two orthogonal guarantees are in force. Input and linkage privacy are cryptographic: they protect each party’s raw records during the joint computation. Output privacy is statistical: it protects individuals in what $\mathbf{R}$ receives and releases. Throughout, the parties are *honest-but-curious*: they follow the protocol but analyse every message they receive, including every component of every ciphertext they can decrypt.

Definition 3.1 (Input privacy — semi-honest model). A protocol Π between $\mathbf{R}$ and $\mathbf{O}$ satisfies *input privacy* in the semi-honest model if, for each party $P \in \{\mathbf{R}, \mathbf{O}\}$, the view of P during the execution of Π is computationally indistinguishable from a distribution that can be simulated using only P ’s own private input and the agreed-upon output.

Table 1: Notation

Symbol	Meaning
n_R, n_O	Number of records held by R, by O; $n_O \geq n_R$
n	Number of matched records after linkage; $n \leq n_R$
$d_R, d_O; p$	Number of features held by R, by O; $p = d_R + d_O$
$\mathbf{X}_R \in \mathbb{R}^{n_R \times d_R}$	R's feature matrix (standardized and clipped, Alg. 1)
$\mathbf{X}_O \in \mathbb{R}^{n_O \times d_O}, \mathbf{y} \in \mathbb{R}^{n_O}$	O's feature matrix and response
$\tau \in \text{Sym}(n_O)$	O's secret permutation of its rows; all O indices below are τ -positions
$I \subseteq [n_R]$	Index set of R's matched records
$\pi : I \rightarrow [n_O]$	Alignment: R's matched record k is O's row $\pi(k)$
$\mathbf{b} \in \{0, 1\}^{n_O}$	Selection vector: $b_j = 1$ iff $j \in \pi(I)$
$\mathbf{X}_R^I, \mathbf{X}_O^I, \mathbf{y}^I$	Matched rows of $\mathbf{X}_R, \mathbf{X}_O, \mathbf{y}$ (aligned by π)
$\mathbf{Z} = [\mathbf{X}_R^I \mid \mathbf{X}_O^I]$	Combined feature matrix, $\mathbf{Z} \in \mathbb{R}^{n \times p}$
$\dot{\mathbf{X}}_R \in \mathbb{R}^{n_O \times d_R}$	R's features aligned to O's order: row $\pi(k)$ is $\mathbf{x}_{R,k}^\top$ for $k \in I$, zero elsewhere
B_R, B_O, B_y	Committed bounds on $\ \mathbf{x}_{R,k}\ _2, \ \mathbf{x}_{O,i}\ _2, y_i $
σ	Standard deviation of the DP noise
$(pk_B, sk_B), (pk_C, sk_C)$	R's BFV key pair (linkage) and CKKS key pair (aggregation)
$\text{Enc}(\cdot), \boxplus$	Encryption under R's public key; homomorphic addition

Definition 3.2 (Linkage privacy). A protocol satisfies *linkage privacy for O* if O's view is computationally indistinguishable regardless of which subset of O's records participates in the intersection.

Definition 3.3 ((ϵ, δ) -differential privacy). A randomised mechanism $\mathcal{M}$ is (ϵ, δ) -DP if for every pair of adjacent datasets D, D' and every measurable set S , $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$. It is ρ -zero-concentrated DP (ρ -zCDP) if the Rényi divergence of order a between $\mathcal{M}(D)$ and $\mathcal{M}(D')$ is at most ρa for all $a > 1$ [5].

Definition 3.4 (Adjacency). The intersection I (hence n , and which of R's records match) is treated as revealed to R; it is R's legitimate PSI output. Two datasets are adjacent if they differ in a single matched individual's O-side data, i.e. in one pair $(\mathbf{x}_{O,i}, y_i)$ with $\mathbf{x}_R$ held fixed (*replace-one* adjacency). DP protects the matched

individuals' O-attributes and outcomes; linkage privacy protects O's unmatched identities.

Remark 3.1 (What is revealed by design). Under Definition 3.4, R learns which of its cohort appear in O's register, and n . When membership is itself sensitive (e.g. a benefits register, where membership *is* the outcome), this must be disclosed to data stewards. A membership-private variant exists (circuit-PSI so that neither party learns $\mathbf{b}$, with the Gram computed on shares), but then $\mathbf{A}$ depends on O's data and must be noised as well: the adjacency becomes add/remove from O's register, the sensitivity gains a B_R^4 term, and the clean- $\mathbf{A}$ advantage of the present design is lost.

4 The Protocol

4.1 Setup

- R holds: $\mathbf{X}_R$, identifier column ID_R ; R has no access to $\mathbf{y}$.
- O holds: $\mathbf{X}_O$, $\mathbf{y}$, identifier column ID_O .

R is to obtain $\hat{\beta}$ (after DP) without O learning $\mathbf{X}_R$ or the linkage map, and without R learning $\mathbf{X}_O$ or $\mathbf{y}$ beyond what the released statistics imply. Figure 1 gives the message flow; Algorithms 1–5 give pseudocode.

4.2 Protocol

Protocol 4.1 (Revised Scenario-B protocol).

Phase 0 — Setup and commitments (Alg. 1). R standardizes $\mathbf{X}_R$ with its own constants, which never leave R, clips rows to the committed bound B_R , and generates a BFV key pair for linkage and a CKKS key pair for aggregation, both at 128-bit security with modulus headroom for flooding. O standardizes $(\mathbf{X}_O, \mathbf{y})$ with *public* constants and clips to B_O and B_y , computes the replace-one sensitivity Δ (Proposition 5.1) and σ , and refuses the run if its privacy ledger would exceed the cap. Only public constants cross in either direction.

Phase 1 — Private record linkage (Alg. 2, Appendix B). O applies a secret permutation τ to its rows. The parties run labeled PSI, in which R sends encrypted windowed powers of its keyed-hashed identifiers and O returns masked, sanitized evaluations of its membership and label polynomials, padded to public degrees. R obtains I and π , the labels being uniformly random τ -positions; O learns only public sizes. R builds $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ and computes

$$\mathbf{A} = \dot{\mathbf{X}}_R^\top \dot{\mathbf{X}}_R = \sum_{k \in I} \mathbf{x}_{R,k} \mathbf{x}_{R,k}^\top \in \mathbb{R}^{d_R \times d_R}$$

locally; $\mathbf{A}$ is retained through to the solve and never transmitted.

Scenario B, revised: sanitized HE, replace-one DP, privacy ledger

honest-but-curious parties · orange = new or changed vs. the draft protocol · numbered badges = risk fixed (key at bottom)

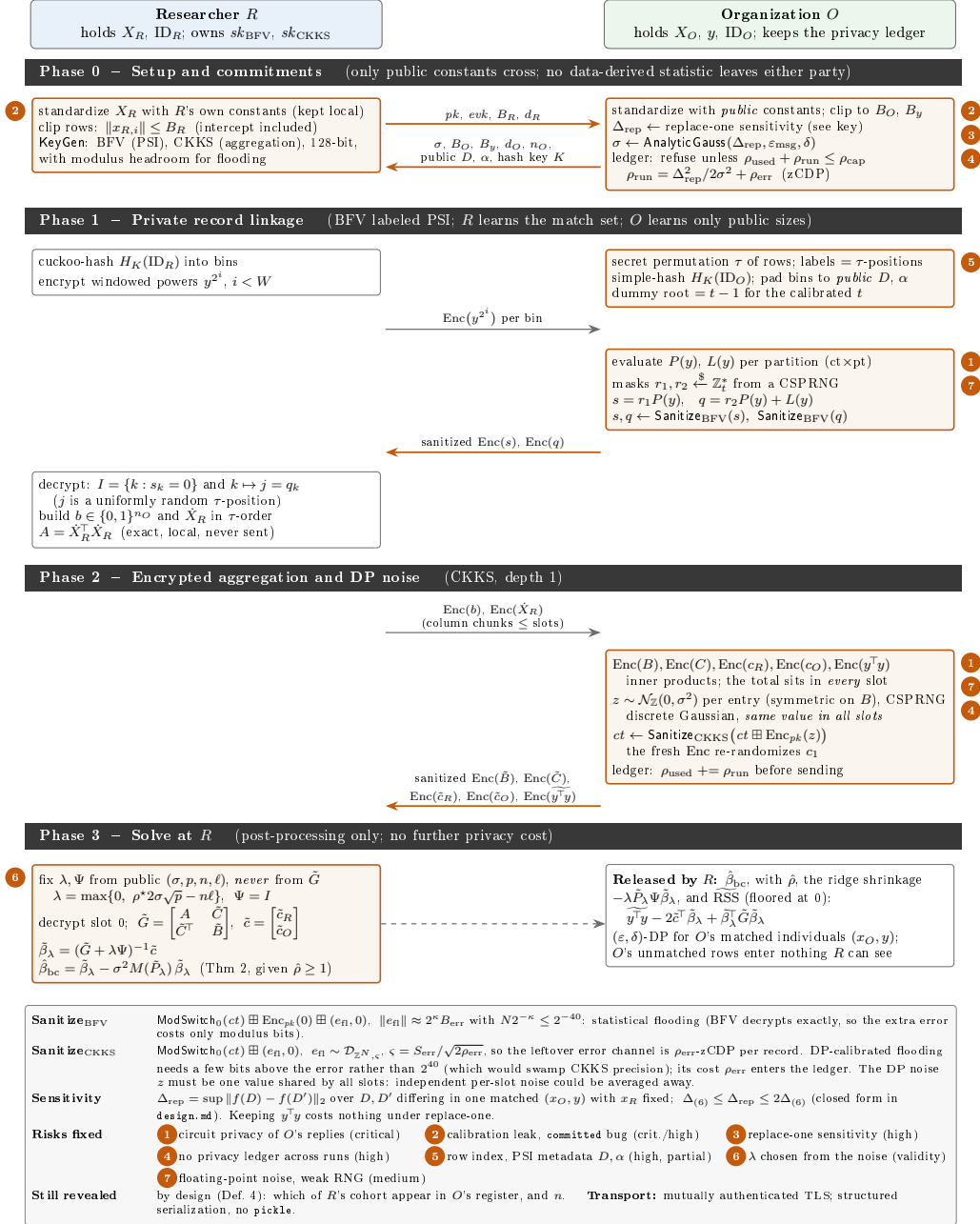


Figure 1: Message flow of Protocol 4.1. Orange marks steps that are new or changed relative to the earlier specification; numbered badges refer to the risks listed in the figure's key.

Phase 2 — Encrypted aggregation and DP noise (Alg. 3). R sends $\text{Enc}(\mathbf{b})$ and $\text{Enc}(\dot{\mathbf{X}}_R)$ in column chunks of at most the slot count. Using plaintext $\mathbf{X}_O$ and $\mathbf{y}$, all at multiplicative depth one, O computes

$$\text{Enc}(B_{jk}) = \sum_i \text{Enc}(b_i) X_{O,ij} X_{O,ik}, \quad \text{Enc}(C_{jk}) = \sum_i \text{Enc}(\dot{X}_{R,ij}) X_{O,ik}, \quad (3)$$

$$\text{Enc}(c_{O,k}) = \sum_i \text{Enc}(b_i) X_{O,ik} y_i, \quad \text{Enc}(c_{R,j}) = \sum_i \text{Enc}(\dot{X}_{R,ij}) y_i, \quad (4)$$

and $\text{Enc}(\mathbf{y}^\top \mathbf{y}) = \sum_i \text{Enc}(b_i) y_i^2$, each inner product leaving the total in every slot. (C needs no $\mathbf{b}$, since $\dot{\mathbf{X}}_R$ is already zero off the matched set; $\mathbf{c}_R$ is the central cross-term, in which O multiplies R's encrypted features by its own plaintext outcome.) For every released entry O draws discrete-Gaussian noise $z \sim \mathcal{N}_{\mathbb{Z}/\gamma}(0, \sigma^2)$ (symmetric on $\mathbf{B}$), adds a *fresh encryption* of z carried identically in every slot, and sanitizes the result (Section 6). O records the run in its ledger before sending.

Phase 3 — Solve at R (Alg. 4). R fixes the ridge (λ, Ψ) from public inputs *before* decrypting (Section 7.5), decrypts slot 0 of each ciphertext, splices in its exact local $\mathbf{A}$,

$$\tilde{\mathbf{G}} = \begin{bmatrix} \mathbf{A} & \tilde{\mathbf{C}} \\ \tilde{\mathbf{C}}^\top & \tilde{\mathbf{B}} \end{bmatrix}, \quad \tilde{\mathbf{c}} = \begin{bmatrix} \tilde{\mathbf{c}}_R \\ \tilde{\mathbf{c}}_O \end{bmatrix},$$

solves $\tilde{\boldsymbol{\beta}}_\lambda = (\tilde{\mathbf{G}} + \lambda \Psi)^{-1} \tilde{\mathbf{c}}$, and applies the zero-budget bias correction of Section 7.6. It releases $\tilde{\boldsymbol{\beta}}_{\text{bc}}$ together with the validity statistic $\hat{\rho}$, the deterministic ridge shrinkage, and $\widetilde{\text{RSS}}$.

4.3 Direction-of-data ledger

R $\rightarrow$ O	pk_B, pk_C , evaluation keys; B_R, d_R, n_R ; BFV encryptions of windowed powers of hashed identifiers; $\text{Enc}(\mathbf{b}), \text{Enc}(\dot{\mathbf{X}}_R)$
O $\rightarrow$ R	$\sigma, B_O, B_y, d_O, n_O$, public (D, α) , hash key K , public standardization constants; sanitized PSI replies; sanitized $\text{Enc}(\tilde{\mathbf{B}}), \text{Enc}(\tilde{\mathbf{C}}), \text{Enc}(\tilde{\mathbf{c}}_R), \text{Enc}(\tilde{\mathbf{c}}_O), \text{Enc}(\widetilde{\mathbf{y}^\top \mathbf{y}})$
never transmitted	$\mathbf{A}$ (local to R); $\mathbf{y}$ (local to O); sk_B, sk_C ; raw $\mathbf{X}_R, \mathbf{X}_O$; τ ; the noise draws; any statistic computed from either party's data other than the released noisy aggregates

Table 2: What crosses between the parties. All messages travel over a mutually authenticated TLS channel with structured (non-executable) serialization.

Under Definition 3.4, $\mathbf{A}$ is a deterministic function of $(I, \mathbf{X}_R)$, both known to R; it carries no incremental information about O and correctly receives no noise.

- Remark 4.1** (Corrections relative to the earlier specification). (i) R no longer uploads $\text{Enc}(\mathbf{A})$. O never consumes $\mathbf{A}$; the final solve occurs at R, which forms $\mathbf{A}$ from its own records the moment linkage completes. Keeping $\mathbf{A}$ local is lossless, and it is what makes the top-left block of the Gram noise vanish (Section 7).
- (ii) The DP noise is added by a fresh encryption followed by sanitization, not by a plaintext-scalar addition (Section 6).
 - (iii) The noise is calibrated to the replace-one sensitivity that Definition 3.4 requires, not to the add/remove bound (Section 5.2).
 - (iv) Calibration is split between the parties: no statistic derived from either party’s data (quantiles, clip rates, sample means or standard deviations) is exchanged.
 - (v) O keeps a privacy ledger, so that re-runs and retraining compose under an explicit cap.
 - (vi) The ridge is fixed from public inputs rather than chosen from the noisy Gram (Section 7.5).
 - (vii) The linkage uses labeled PSI with data-independent labels and metadata (Appendix B).

5 Output Privacy: DP Noise on Encrypted Aggregates

5.1 The noise injection mechanism

For every released entry s of the stacked statistic

$$\mathbf{f}(D) = (\text{triu}(\mathbf{B}), \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top \mathbf{y}), \quad (5)$$

O draws $z_s \sim \mathcal{N}_{\mathbb{Z}/\gamma}(0, \sigma^2)$ independently (a discrete Gaussian on the grid $\gamma^{-1}\mathbb{Z}$ at the CKKS resolution [16]), mirrors the draws for $\mathbf{B}$ across the diagonal, and sets

$$ct_s \leftarrow \text{SANITIZECKKS}(\text{Enc}(s) \boxplus \text{Enc}_{pk_C}(z_s \cdot \mathbf{1})). \quad (6)$$

O retains z_s and never discloses it; R decrypts $\tilde{s} = s + z_s$ (up to a CKKS perturbation far below σ) and cannot decompose it. Two details matter:

- *The same z_s in every slot.* The inner products of (3)–(4) leave the total in all $N_C/2$ slots. Independent noise per slot would let R average the copies and remove the noise; the constant vector $z_s \mathbf{1}$ gives every copy the same noise.
- *A fresh encryption, then sanitization.* A plaintext-scalar addition would not re-randomize the ciphertext; Section 6 shows this is fatal.

Because σ is a public constant chosen by O, the privacy budget stays under O’s control, and every function of the released statistics computed by R from its own data inherits the guarantee by post-processing.

Algorithm 1 SETUP: commitments and calibration (Phase 0)

Input: R: $\mathbf{X}_R, ID_R$. O: $\mathbf{X}_O, \mathbf{y}, ID_O$, public standardization constants $(\boldsymbol{\mu}_O, \mathbf{s}_O, \mu_y, s_y)$, privacy ledger $\mathcal{L}$, cap ρ_{cap} . Public: $(\varepsilon_{\text{msg}}, \delta)$, ρ_{err} , HE parameters, PSI parameters $(N_{\text{bins}}, h, d_{\text{cap}}, W)$, flooding parameter κ .

Output: Public parameter set $\mathcal{P}$; R holds (sk_B, sk_C) .

At R:

- 1: Standardize $\mathbf{X}_R$ with R's own constants, which never leave R; append the intercept column if one is used
- 2: $\mathbf{x}_{R,k} \leftarrow \mathbf{x}_{R,k} \cdot \min\{1, B_R / \|\mathbf{x}_{R,k}\|_2\}$ for all $k \in [n_R]$ $\triangleright$ enforce the committed bound
- 3: $(pk_B, sk_B, evk_B) \leftarrow \text{BFV.KeyGen}(N_B, t, q_B);$ $(pk_C, sk_C, evk_C) \leftarrow \text{CKKS.KeyGen}(N_C, \Delta_C, q_C)$ $\triangleright$ 128-bit; q includes flooding headroom
- 4: **send** $(pk_B, evk_B, pk_C, evk_C, B_R, d_R, n_R)$ **to** O

At O:

- 5: Standardize $(\mathbf{X}_O, \mathbf{y})$ with the public constants; clip $\|\mathbf{x}_{O,i}\|_2 \leq B_O$ and $|y_i| \leq B_y$ for all $i \in [n_O]$
 - 6: $\Delta \leftarrow \text{DELTA REP}(B_R, B_O, B_y)$ $\triangleright$ replace-one sensitivity, Alg. 5
 - 7: $\sigma \leftarrow \text{ANALYTIC GAUSS}(\Delta, \varepsilon_{\text{msg}}, \delta);$ $\rho_{\text{run}} \leftarrow \Delta^2 / (2\sigma^2) + \rho_{\text{err}}$
 - 8: **if** $\mathcal{L}[\text{R}, \text{dataset}] + \rho_{\text{run}} > \rho_{\text{cap}}$ **then**
 - 9: **abort** $\triangleright$ budget exhausted
 - 10: **end if**
 - 11: $K \xleftarrow{\$} \{0, 1\}^{256};$ $(D, \alpha) \leftarrow$ public degree and partition count computed from $(n_O, h, N_{\text{bins}}, d_{\text{cap}})$
 - 12: **send** $\mathcal{P} = (\sigma, B_O, B_y, d_O, n_O, D, \alpha, K, \boldsymbol{\mu}_O, \mathbf{s}_O, \mu_y, s_y)$ **to** R
-

Algorithm 2 LINK: labeled PSI with record alignment (Phase 1)

Input: R: $ID_R, \mathbf{X}_R, sk_B$. O: ID_O , rows $(\mathbf{x}_{O,i}, y_i)_{i \in [n_O]}$. Public: K, t, N_{bins} , hash functions $\beta_1, \dots, \beta_h, D, \alpha, W$.

Output: R: $I, \pi, \mathbf{b}, \dot{\mathbf{X}}_R, \mathbf{A}$. O: nothing.

At O: ▷ offline preprocessing

- 1: $\tau \xleftarrow{\$} \text{Sym}(n_O)$; re-index the rows of $(ID_O, \mathbf{X}_O, \mathbf{y})$ by τ ▷ uniform labels
- 2: $v_j \leftarrow H_K(ID_{O,j}) \bmod (t-1)$ for $j \in [n_O]$ ▷ never equals the dummy root $t-1$
- 3: Insert (v_j, j) into bins $\beta_1(v_j), \dots, \beta_h(v_j)$ ▷ simple hashing
- 4: **if** a bin holds more than αD items, or two items in one bin share a value v **then**
- 5: **abort** ▷ probability $\leq \delta_{\text{psi}}$
- 6: **end if**
- 7: **for** each bin $\beta \in [N_{\text{bins}}]$ and partition $a \in [\alpha]$ **do**
- 8: $S_{\beta,a} \leftarrow$ the a -th block of at most D items in bin β
- 9: $P_{\beta,a}(X) \leftarrow \prod_{(v,j) \in S_{\beta,a}} (X-v) \cdot (X-(t-1))^{D-|S_{\beta,a}|}$ ▷ padded to the public degree D
- 10: $L_{\beta,a} \leftarrow$ the interpolant of degree $< D$ with $L_{\beta,a}(v) = j$ for all $(v, j) \in S_{\beta,a}$
- 11: **end for**

At R:

- 12: $u_k \leftarrow H_K(ID_{R,k}) \bmod (t-1)$; cuckoo-insert every u_k into N_{bins} bins using $\beta_1, \dots, \beta_h$;
 let $T[\beta] = k$ for occupied bins ▷ batches if n_R exceeds capacity
- 13: $w_\beta \leftarrow u_{T[\beta]}$ for occupied bins, $w_\beta \leftarrow 0$ otherwise
- 14: **send** $\text{BFV.Enc}_{pk_B}(\mathbf{w}^{\odot 2^i})$ for $i = 0, \dots, W-1$ **to O**

At O:

- 15: $ct_d \leftarrow \text{Enc}(\mathbf{w}^{\odot d})$ for $d = 1, \dots, D$, as products of window ciphertexts
- 16: **for** $a = 1, \dots, \alpha$ **do**
- 17: $ct_P \leftarrow \mathbf{p}_{a,0} \boxplus \boxplus_{d=1}^D \mathbf{p}_{a,d} \odot ct_d$; $ct_L \leftarrow \ell_{a,0} \boxplus \boxplus_{d=1}^{D-1} \ell_{a,d} \odot ct_d$ ▷ also valid for $D = 1$
- 18: $\mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} (\mathbb{Z}_t^*)^{N_{\text{bins}}}$ from a CSPRNG
- 19: $ct_s \leftarrow \text{SANITIZEBFV}(\mathbf{r}_1 \odot ct_P)$; $ct_q \leftarrow \text{SANITIZEBFV}(\mathbf{r}_2 \odot ct_P \boxplus ct_L)$
- 20: **send** (ct_s, ct_q) **to R**
- 21: **end for**

At R:

- 22: **for** each occupied bin β with $k = T[\beta]$, and each $a \in [\alpha]$ **do**
- 23: $s \leftarrow \text{Dec}_{sk_B}(ct_s^{(a)})_\beta$; $q \leftarrow \text{Dec}_{sk_B}(ct_q^{(a)})_\beta$
- 24: **if** $s = 0$ **and** $0 \leq q < n_O$ **then**
- 25: $\pi(k) \leftarrow q$
- 26: **end if**
- 27: **end for**
- 28: $I \leftarrow \text{dom}(\pi)$; $\mathbf{b} \leftarrow \mathbf{0}_{n_O}$; $\dot{\mathbf{X}}_R \leftarrow \mathbf{0}_{n_O \times d_R}$
- 29: **for** $k \in I$: $b_{\pi(k)} \leftarrow 1$ and $\dot{\mathbf{X}}_R[\pi(k), :] \leftarrow \mathbf{x}_{R,k}^\top$
- 30: $\mathbf{A} \leftarrow \dot{\mathbf{X}}_R^\top \dot{\mathbf{X}}_R$ ▷ exact; never transmitted

Algorithm 3 AGGREGATE: encrypted sufficient statistics with DP noise (Phase 2)

Input: R: $\mathbf{b}, \dot{\mathbf{X}}_R, pk_C$. O: $\mathbf{X}_O, \mathbf{y}$ in τ -order, $\sigma, \rho_{\text{run}}$, ledger $\mathcal{L}$. Public: slot count S .

Output: R receives sanitized encryptions of $\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{c}}_R, \tilde{\mathbf{c}}_O, \mathbf{y}^\top \mathbf{y}$.

At R:

- 1: Partition $[n_O]$ into chunks $\mathcal{C}_1, \dots, \mathcal{C}_m$ of at most S positions
- 2: **send** CKKS.Enc($\mathbf{b}_{\mathcal{C}_c}$) and CKKS.Enc($\dot{\mathbf{X}}_R[\mathcal{C}_c, j]$) for all $c \in [m], j \in [d_R]$ **to** O

At O:

- 3: $\mathcal{E} \leftarrow \{(\mathbf{b}, \mathbf{x}_{O,j} \odot \mathbf{x}_{O,k}) : j \leq k\} \cup \{(\dot{\mathbf{X}}_R[:, j], \mathbf{x}_{O,k})\} \cup \{(\dot{\mathbf{X}}_R[:, j], \mathbf{y})\} \cup \{(\mathbf{b}, \mathbf{x}_{O,k} \odot \mathbf{y})\} \cup \{(\mathbf{b}, \mathbf{y} \odot \mathbf{y})\}$
 $\triangleright$ (encrypted, plaintext) pairs giving $B_{jk}, C_{jk}, c_{R,j}, c_{O,k}, \mathbf{y}^\top \mathbf{y}$
 - 4: **for** each $(\mathbf{u}, \mathbf{w}) \in \mathcal{E}$ **do**
 - 5: $ct \leftarrow \boxplus_{c=1}^m \text{EvalSum}(\text{Enc}(\mathbf{u}_{\mathcal{C}_c}) \odot \mathbf{w}_{\mathcal{C}_c})$ $\triangleright$ depth 1; the total occupies every slot
 - 6: $z \leftarrow \mathcal{N}_{\mathbb{Z}/\gamma}(0, \sigma^2)$ from a CSPRNG $\triangleright$ discrete Gaussian at resolution $1/\gamma$; one draw per released entry
 - 7: $ct \leftarrow \text{SANITIZECKKS}(ct \boxplus \text{CKKS.Enc}_{pk_C}(z \cdot \mathbf{1}))$ $\triangleright$ same z in every slot; fresh c_1
 - 8: **end for**
 - 9: $\mathcal{L}[\text{R}, \text{dataset}] \leftarrow \mathcal{L}[\text{R}, \text{dataset}] + \rho_{\text{run}}$ $\triangleright$ recorded before release
 - 10: **send** all sanitized ciphertexts **to** R
-

Algorithm 4 SOLVE: assembly, fixed ridge and bias correction at R (Phase 3)

Input: R: $\mathbf{A}, sk_C$, the ciphertexts of Alg. 3. Public: $\sigma, p = d_R + d_O, n = |I|$, validity target ρ^* , committed eigenvalue floor $\ell \geq 0$.

Output: $\hat{\beta}_{\text{bc}}$, the validity statistic $\hat{\rho}$, the ridge shrinkage $\mathbf{s}_\lambda$, and $\widetilde{\text{RSS}}$.

- 1: $\lambda \leftarrow \max\{0, 2\rho^*\sigma\sqrt{p} - n\ell\}$; $\Psi \leftarrow \mathbf{I}_p$ $\triangleright$ fixed before decryption, never from $\tilde{\mathbf{G}}$
 - 2: Decrypt slot 0 of every ciphertext; set $\tilde{B}_{kj} \leftarrow \tilde{B}_{jk}$ for $j < k$
 - 3: $\tilde{\mathbf{G}} \leftarrow \begin{bmatrix} \mathbf{A} & \tilde{\mathbf{C}} \\ \tilde{\mathbf{C}}^\top & \tilde{\mathbf{B}} \end{bmatrix}$; $\tilde{\mathbf{c}} \leftarrow \begin{bmatrix} \tilde{\mathbf{c}}_R \\ \tilde{\mathbf{c}}_O \end{bmatrix}$
 - 4: $\hat{\rho} \leftarrow \lambda_{\min}(\tilde{\mathbf{G}} + \lambda\Psi)/(2\sigma\sqrt{p})$
 - 5: **if** $\hat{\rho} < 1$ **then**
 - 6: report that the validity check failed and withhold $\hat{\beta}$ $\triangleright$ post-processing; no privacy cost
 - 7: **end if**
 - 8: $\tilde{\mathbf{P}} \leftarrow (\tilde{\mathbf{G}} + \lambda\Psi)^{-1}$; $\tilde{\beta} \leftarrow \tilde{\mathbf{P}}\tilde{\mathbf{c}}$
 - 9: $\hat{\beta}_{\text{bc}} \leftarrow \tilde{\beta} - \sigma^2 \text{BIASOP}(\tilde{\mathbf{P}}, d_R) \tilde{\beta}$ $\triangleright$ Theorem 7.2
 - 10: $\mathbf{s}_\lambda \leftarrow -\lambda\tilde{\mathbf{P}}\Psi\tilde{\beta}$; $\widetilde{\text{RSS}} \leftarrow \max\{0, \mathbf{y}^\top \mathbf{y} - 2\tilde{\mathbf{c}}^\top \tilde{\beta} + \tilde{\beta}^\top \tilde{\mathbf{G}} \tilde{\beta}\}$
 - 11: **return** $\hat{\beta}_{\text{bc}}, \hat{\rho}, \mathbf{s}_\lambda, \widetilde{\text{RSS}}$, back-transformed with R's constants and O's public constants
-

Algorithm 5 Subroutines

```

1: function SANITIZEBFV( $ct$ )                                 $\triangleright$  statistical flooding; BFV decryption stays exact
2:    $ct \leftarrow \text{ModSwitch}_0(ct)$ 
3:    $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^{N_B}, 2^\kappa B_{\text{err}}}$                                  $\triangleright B_{\text{err}}$ : bound on the error of  $ct$  at level 0
4:   return  $ct \boxplus \text{BFV.Enc}_{pk_B}(\mathbf{0}) \boxplus (\mathbf{e}, 0)$ 
5: end function

6: function SANITIZECKKS( $ct$ )                                 $\triangleright$  DP-calibrated flooding of the error channel
7:    $ct \leftarrow \text{ModSwitch}_0(ct)$ 
8:    $\varsigma \leftarrow S_{\text{err}}/\sqrt{2\rho_{\text{err}}}$ ;  $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^{N_C}, \varsigma}$                                  $\triangleright$  Lemma 6.1
9:   return  $ct \boxplus (\mathbf{e}, 0)$ 
10: end function

11: function DELTAREP( $B_R, B_O, B_y$ )                                 $\triangleright$  Proposition 5.1
12:    $R \leftarrow B_R^2$ ,  $O \leftarrow B_O^2$ ,  $Y \leftarrow B_y^2$ 
13:   if  $|R - Y| \leq 2O$  then
14:     return  $(\frac{1}{2}(2O + R + Y)^2 + 2RY)^{1/2}$ 
15:   else if  $R - Y \geq 2O$  then
16:     return  $(4R(O + Y))^{1/2}$ 
17:   else
18:     return  $(\max_{t \in [-B_y, B_y]} F(c^*(t), t))^{1/2}$                                  $\triangleright F, c^*$  as in (9)
19:   end if
20: end function

21: function ANALYTICGAUSS( $\Delta, \varepsilon, \delta$ )                                 $\triangleright$  Balle–Wang
22:   return the smallest  $\sigma$  with  $\Phi(\frac{\Delta}{2\sigma} - \frac{\varepsilon\sigma}{\Delta}) - e^\varepsilon \Phi(-\frac{\Delta}{2\sigma} - \frac{\varepsilon\sigma}{\Delta}) \leq \delta$                                  $\triangleright$  bisection
23: end function

24: function BIASOP( $\mathbf{P}, d_R$ )                                 $\triangleright$  bracket of Theorem 7.1
25:    $\Pi_R \leftarrow \text{diag}(\mathbf{1}_{d_R}, \mathbf{0}_{d_O})$ ;  $\Pi_O \leftarrow \mathbf{I}_p - \Pi_R$ 
26:   return  $(\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P} - \mathbf{P} \text{diag}(\mathbf{P}))\Pi_O + (\mathbf{P}\Pi_O\mathbf{P} + \text{tr}(\Pi_O\mathbf{P})\mathbf{P})\Pi_R$ 
27: end function

```

5.2 Sensitivity under replace-one adjacency

A subtlety arises for B_R : because $\mathbf{X}_R$ is encrypted, $\mathbf{O}$ cannot compute $\|\mathbf{x}_{R,k}\|_2$ directly. $\mathbf{R}$ must commit to B_R during setup, before any data is exchanged, and enforce it by clipping; this allows $\mathbf{O}$ to calibrate the noise without seeing the data. If $\mathbf{R}$ appends an intercept column, the committed bound is $\sqrt{1 + B_{R,\text{rows}}^2}$.

Replacing one matched individual's $(\mathbf{x}, y)$ by $(\mathbf{x}', y')$ with $\mathbf{x}_R$ fixed changes the released vector by

$$(\text{triu}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top), \mathbf{x}_R(\mathbf{x} - \mathbf{x}')^\top, \mathbf{x}_R(y - y'), \mathbf{x}y - \mathbf{x}'y', y^2 - y'^2).$$

Let $\Delta = \sup \|\mathbf{f}(D) - \mathbf{f}(D')\|_2$ over adjacent D, D' , and let

$$\Delta_{\text{add}}^2 = (B_O^2 + B_R^2)(B_O^2 + B_y^2) + B_y^4 \quad (7)$$

be the bound obtained when one record is added or removed. The earlier specification used (7), which does not match Definition 3.4.

Proposition 5.1 (Replace-one sensitivity). $\Delta_{\text{add}} \leq \Delta \leq 2\Delta_{\text{add}}$. Moreover, with $R = B_R^2$, $O = B_O^2$, $Y = B_y^2$,

$$\Delta^2 = \begin{cases} \frac{1}{2}(2O + R + Y)^2 + 2RY & \text{if } |R - Y| \leq 2O, \\ 4R(O + Y) & \text{if } R - Y \geq 2O, \\ \max_{t \in [-B_y, B_y]} F(c^*(t), t) & \text{if } Y - R > 2O, \end{cases} \quad (8)$$

where

$$\begin{aligned} F(c, t) &= 2O^2(1 - c^2) + 2RO(1 - c) + R(B_y - t)^2 \\ &\quad + O(Y + t^2) - 2OcB_yt + (Y - t^2)^2, \\ c^*(t) &= \min\{1, \max\{-1, -(R + B_yt)/(2O)\}\}. \end{aligned} \quad (9)$$

Proof sketch. The lower bound follows by replacing $(\mathbf{x}, y)$ with $(\mathbf{0}, 0)$, which reproduces the add/remove change. For the upper bound, write $\mathbf{f}(D) - \mathbf{f}(D') = [\mathbf{f}(D) - \mathbf{f}(D^0)] + [\mathbf{f}(D^0) - \mathbf{f}(D')]$, where D^0 has the individual's $\mathbf{O}$ -side data set to zero, and apply the triangle inequality. For (8), the objective is increasing in $\|\mathbf{x}_R\|$, so $\|\mathbf{x}_R\| = B_R$; at the optimum $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$ and $|y| = B_y$, and the remaining dependence is on $c = \cos \angle(\mathbf{x}, \mathbf{x}')$ and $t = y'$. Maximizing the concave quadratic in c gives $c^*(t)$, and the three cases correspond to where the maximizer in t lies. The closed form agrees with brute-force maximization to within 5×10^{-9} relative on fifteen bound settings covering all three regimes (`reports/checks/replace_one_closed_form_check.py`). In the third regime the one-dimensional maximum must be computed with local refinement or rounded up, since σ must never be calibrated to an underestimate. $\square$

Remark 5.1 (Magnitude, and $\mathbf{y}^\top \mathbf{y}$ is free). For $B_R = B_O = B_y$ the ratio $\Delta/\Delta_{\text{add}}$ is $\sqrt{2}$; it approaches 2 when B_R dominates (many $\mathbf{R}$ features, or the intercept) and 1

when B_y dominates. For $B_R = B_O = \sqrt{5}$, $B_y = 5$ it is 31.6/30.4. Releasing $\mathbf{y}^\top \mathbf{y}$ costs nothing under replace-one: the worst case replaces $(\mathbf{x}, y)$ with $(-\mathbf{x}, -y)$, which leaves y^2 unchanged, and Δ is numerically identical with or without $\mathbf{y}^\top \mathbf{y}$ in all settings we checked.

The upper triangle of $\mathbf{B}$, rather than the full matrix, is what is released and noised; using the full Frobenius norm of $\Delta \mathbf{B}$ would be conservative.

5.3 Calibration, composition and the ledger

$\mathcal{O}$ calibrates σ with the analytic Gaussian mechanism of Balle and Wang [19]: the smallest σ with

$$\Phi\left(\frac{\Delta}{2\sigma} - \frac{\varepsilon\sigma}{\Delta}\right) - e^\varepsilon \Phi\left(-\frac{\Delta}{2\sigma} - \frac{\varepsilon\sigma}{\Delta}\right) \leq \delta,$$

evaluating the left tail with $\frac{1}{2}\text{erfc}$ to avoid cancellation. Whatever rule chooses σ , the (discrete) Gaussian mechanism with ℓ_2 -sensitivity Δ is ρ -zCDP with $\rho = \Delta^2/(2\sigma^2)$ [5, 16]; zCDP composes additively and converts to (ε, δ) -DP with $\varepsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$.

$\mathcal{O}$ therefore keeps a ledger per (dataset, researcher). A run costs

$$\rho_{\text{run}} = \frac{\Delta^2}{2\sigma^2} + \rho_{\text{err}}, \quad (10)$$

where ρ_{err} is the cost of the ciphertext-error channel (Lemma 6.1); $\mathcal{O}$ refuses a run if the cumulative spend would exceed its cap ρ_{cap} , and records the spend before releasing. Retraining, re-runs, and the per-query noise of split inference (Section 8.2) are all charged to the same ledger.

5.4 Choice of noise distribution

The noise distribution determines the type of privacy guarantee, the magnitude of noise required, and the correctness of the mechanism under finite-precision arithmetic.

Laplace mechanism. Adding $\mathbf{z} \sim \text{Lap}(\mathbf{0}, \Delta_1/\varepsilon)^d$ to a d -dimensional aggregate with ℓ_1 -sensitivity Δ_1 achieves pure $(\varepsilon, 0)$ -DP. However, $\Delta_1 \leq \sqrt{d} \Delta_2$ with equality when the sensitivity is spread uniformly across coordinates. For our aggregates the ambient dimension is of order d_O^2 or $d_R d_O$, so Laplace noise can be 10 to 50 times larger per entry than Gaussian noise at the same ε .

Gaussian mechanism. Adding $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ with $\sigma \geq \Delta_2 \sqrt{2 \ln(1.25/\delta)}/\varepsilon$ (valid for $\varepsilon < 1$) achieves (ε, δ) -DP. Calibrated to Δ_2 , it adds far less noise per coordinate for high-dimensional outputs; the analytic calibration above gives the smallest valid σ .

Discrete Gaussian (used here). Floating-point samplers leak through their low-order bits [14], and CKKS encodes values on a grid. We therefore draw from the discrete Gaussian $\mathcal{N}_{\mathbb{Z}/\gamma}(0, \sigma^2)$ with a cryptographically secure generator; it enjoys the same zCDP guarantee as the continuous Gaussian [16]. The interaction between the grid, the CKKS approximation error and the sensitivity is part of Proof obligation 10.1.

Remark 5.2 (What the guarantee covers). The released statistics, and hence $\hat{\beta}_{bc}$, are DP with respect to the matched individuals’ O-attributes and outcomes. O’s unmatched rows do not enter the decrypted values (they are masked by $\mathbf{b}$ and by the zeros of $\dot{\mathbf{X}}_R$), and they are covered in the ciphertext-error channel by Lemma 6.1. DP does *not* protect membership or n (Remark 3.1), nor R’s own cohort in a published model, since $\mathbf{A}$ is exact. Quantities R computes from the release and its own data — for instance $\mathbf{x}_{R,k}^\top \hat{\beta}_R$ — are post-processing.

6 Circuit Privacy: Sanitizing O’s Replies

The DP argument of Section 5 reasons about the decrypted value $s + z$. But R receives, and can fully decrypt, a ciphertext that O computed from its plaintext data. Unless that ciphertext is sanitized, it can reveal far more than $s + z$.

6.1 Why plaintext-scalar noise fails

A CKKS ciphertext is a pair (c_0, c_1) with $c_0 + c_1 s_{\text{key}} \approx \Delta_C m$. Adding a plaintext constant changes only c_0 . Multiplying by a plaintext, rotating (key-switching with R’s evaluation keys), and adding produce a c_1 that is a deterministic function of R’s own ciphertexts, R’s evaluation keys, and O’s plaintext columns — and of nothing else. In particular c_1 does not depend on the noise z .

A DP adversary knows every record but one. R can therefore recompute c_1 for each candidate value of the unknown record and compare with the reply. In a real OpenFHE run of the earlier specification’s noise step (plaintext-scalar addition), R identified the true dataset in 20 of 20 trials, both for a matched record and for an *unmatched* one; an $\epsilon = 1$ guarantee would cap the success rate at $e/(1 + e) \approx 73\%$. Unmatched records leak because $\mathbf{b}$ masks them only in the message: the c_1 component of $\text{Enc}(\mathbf{b})$ is random in every slot, so every row of $\mathbf{X}_O$ and $\mathbf{y}$ enters c_1 . The same reasoning applies to O’s BFV linkage replies and, even more directly, to split inference (Section 8.2).

6.2 Sanitization

Fresh encryption. Adding $\text{Enc}_{pk_C}(z\mathbf{1})$ with fresh randomness makes c_1 of the reply computationally uniform and independent of O’s data (by RLWE, since the encryption randomness is O’s). What remains is the decryption error $e_{\text{data}} + e_{\text{fresh}}$, and e_{data} still

depends on O 's plaintext columns through R 's known encryption and key-switching errors.

Flooding the error. Classical statistical flooding adds an error 2^κ times larger than e_{data} . For BFV this is harmless, because decryption rounds the error away, and costs only modulus bits. For CKKS the error lands in the decoded value: flooding by $2^{40}\|e_{\text{data}}\|$ would require a scale beyond 2^{60} to preserve precision. We therefore flood the CKKS error channel with a discrete Gaussian calibrated as a DP mechanism, in the spirit of Li, Micciancio, Schultz and Sorrell [12], which needs only a few bits above the error.

Lemma 6.1 (DP-calibrated flooding). *Let $e_{\text{data}}(D) \in \mathbb{Z}^{N_c}$ be the data-dependent part of the decryption error of a reply after ModSwitch_0 , and let $S_{\text{err}} \geq \sup \|e_{\text{data}}(D) - e_{\text{data}}(D')\|_2$ over adjacent D, D' , where adjacency here includes changing one unmatched record. Adding $\mathbf{e} \sim \mathcal{D}_{\mathbb{Z}^{N_c}, \varsigma}$ with $\varsigma = S_{\text{err}}/\sqrt{2\rho_{\text{err}}}$ makes the error channel $\rho_{\text{err}}\text{-zCDP}$. Together with the fresh encryption, R 's view of each reply is then $(\rho_s + \rho_{\text{err}})\text{-zCDP}$, where ρ_s is the cost of the value channel.*

Proof sketch. The error channel is a discrete Gaussian mechanism applied to e_{data} , with sensitivity S_{err} [16]. c_1 is independent of the data by the fresh encryption. The two channels compose. $\square$

Proof obligation 6.1 (Bounding S_{err}). An explicit bound on S_{err} for the depth-one circuit of (3)–(4) is required. One record enters e_{data} through products of R 's fresh encryption error with the encoding of that record's contribution, followed by rotation and key-switching and the rescale. A bound of the form $S_{\text{err}} \leq c \|e_{\text{in}}\|_\infty \|\text{Ecd}(\Delta\mathbf{w})\|_1 + (\text{data-independent terms})$, holding with probability $1 - \delta_{\text{err}}$ over R 's encryption randomness (charged to δ), is expected; it must be derived for the exact encoding and key-switching used.

Remark 6.1 (Order of operations and cost). Sanitization first switches to the last modulus level, which scales down e_{data} , and then adds the flooding term at that level. For CKKS the extra cost is a few modulus bits and ρ_{err} in the ledger. For BFV, statistical flooding with $N_B 2^{-\kappa} \leq 2^{-40}$ costs about $\kappa + \log_2 B_{\text{err}}$ modulus bits; ring dimension 16384 still suffices for plaintext moduli up to about 2^{40} , and larger moduli need 32768. Where strict simulation (Definition 3.1) is preferred over a DP-protected error channel, CKKS can use statistical flooding with a scale of about $2^{\kappa + \log_2 \|e_{\text{data}}\| + 10}$, at the price of a 128-bit arithmetic backend or composite scaling.

6.3 Input privacy of the revised protocol

Proposition 6.1 (Views, semi-honest). *Assume RLWE, 128-bit parameters, statistical flooding of the BFV replies, and Lemma 6.1.*

- (a) $\mathcal{O}$'s view consists of public constants and ciphertexts under $\mathcal{R}$'s keys; it is simulatable from public sizes. In particular Definitions 3.1 and 3.2 hold for $\mathcal{R}$'s data and the linkage.
- (b) $\mathcal{R}$'s linkage view is statistically close (distance $\leq 2^{-40}$ per reply) to a simulation from (I, π) : after flooding, each reply depends only on its decrypted values, which are zero or a uniform τ -position on matches and uniform elsewhere, and (D, α) are public.
- (c) $\mathcal{R}$'s aggregation view is ρ_{run} -zCDP with respect to one $\mathcal{O}$ record: c_1 is computationally uniform, the decrypted value is the Gaussian mechanism of Section 5, and the error channel is covered by Lemma 6.1.

Proof sketch. (a) is IND-CPA. (b) follows from the smudging lemma for BFV noise and the choice of masks r_1, r_2 uniform in $\mathbb{Z}_t^*$. (c) combines the fresh-encryption argument, Section 5, Lemma 6.1, and zCDP composition. $\square$

With DP-calibrated flooding, $\mathcal{R}$'s view of $\mathcal{O}$ is protected in the DP sense rather than being strictly simulatable from the output; Definition 3.1 holds for $\mathcal{O}$'s data in this relaxed form.

7 Bias of the Private Estimator

7.1 Noise model

Write the assembled private statistics as $\tilde{\mathbf{G}} = \mathbf{G} + \mathbf{E}$ and $\tilde{\mathbf{c}} = \mathbf{c} + \mathbf{f}$, with

$$\mathbf{E} = \begin{bmatrix} \mathbf{0} & \mathbf{E}_C \\ \mathbf{E}_C^\top & \mathbf{E}_B \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_R \\ \mathbf{f}_O \end{bmatrix}. \quad (11)$$

The top-left $d_R \times d_R$ block of $\mathbf{E}$ is identically zero, since $\mathbf{A}$ is $\mathcal{R}$'s exact local block. Each independent entry on and above the block-masked diagonal is drawn with variance σ^2 and mirrored, so every entry, the diagonal included, has variance σ^2 . With $\chi_{ij} = 0$ iff i, j both lie in the $\mathcal{R}$ block and $\chi_{ij} = 1$ otherwise,

$$\mathbb{E}[E_{ij}E_{kl}] = \sigma^2 \chi_{ij} (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{ik}\delta_{il}). \quad (12)$$

The last term removes the diagonal double-count of the GOE convention, which would put variance $2\sigma^2$ on the diagonal. $\mathbf{E}$ and $\mathbf{f}$ occupy disjoint entries of the release, are independent, and are symmetric about zero. (The discrete Gaussian and the continuous Gaussian share these first and second moments; the CKKS perturbation is of lower order and is ignored here.)

Condition on the realized sample, so $\hat{\beta} = \mathbf{G}^{-1}\mathbf{c}$ is the fixed target, and set $\mathbf{P} = \mathbf{G}^{-1}$,

$$\rho = \frac{\lambda_{\min}(\mathbf{G})}{2\sigma\sqrt{p}}, \quad \mathcal{E}_\eta = \{\|\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}\|_{\text{op}} \leq \eta\} \quad (0 < \eta < 1). \quad (13)$$

$\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$. Define the operator

$$M(\mathbf{P}) = (\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P} - \mathbf{P} \text{diag}(\mathbf{P}))\Pi_O + (\mathbf{P}\Pi_O\mathbf{P} + \text{tr}(\Pi_O\mathbf{P})\mathbf{P})\Pi_R, \quad (14)$$

where $\text{diag}(\mathbf{P})$ is the diagonal matrix formed from the diagonal of $\mathbf{P}$.

Remark 7.1 (The unconditional mean does not exist). With Gaussian $\mathbf{E}$, $\tilde{\boldsymbol{\beta}} = (\mathbf{G} + \mathbf{E})^{-1}(\mathbf{c} + \mathbf{f})$ has no finite mean: already for $p = 1$, $\mathbb{E}[1/(a + e)]$ diverges. Statements about the bias of $\tilde{\boldsymbol{\beta}}$ must therefore be conditional, or about a truncated estimator. We condition on $\mathcal{E}_\eta$, whose complement has exponentially small probability (Remark 7.2).

7.2 Result

Theorem 7.1 (Leading-order conditional bias). *Fix $\eta \in (0, 1)$ with $\eta\rho > 1$. Then*

$$\mathbb{E}[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta] - \hat{\boldsymbol{\beta}} = \sigma^2 M(\mathbf{P}) \hat{\boldsymbol{\beta}} + O(\sigma^4) + O(e^{-p(\eta\rho-1)^2/2}),$$

where the constants depend on $\|\mathbf{P}\|_{\text{op}}$, $\|\hat{\boldsymbol{\beta}}\|$, p and η . Equivalently, splitting the driving coefficients,

$$\begin{aligned} \mathbb{E}[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta] - \hat{\boldsymbol{\beta}} &= \sigma^2 (\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P} - \mathbf{P} \text{diag}(\mathbf{P}))\Pi_O \hat{\boldsymbol{\beta}} \\ &\quad + \sigma^2 (\mathbf{P}\Pi_O\mathbf{P} + \text{tr}(\Pi_O\mathbf{P})\mathbf{P})\Pi_R \hat{\boldsymbol{\beta}} + \dots \end{aligned}$$

The moment-vector noise $\mathbf{f}$ contributes nothing at this order; it is variance-only. The term $-\mathbf{P} \text{diag}(\mathbf{P})\Pi_O$ is the single-draw correction to the GOE contraction.

The proof is given in Appendix A. Setting $\Pi_O \rightarrow \mathbf{I}$ and $\Pi_R \rightarrow \mathbf{0}$ recovers the homogeneous-Gram single-draw bias $\sigma^2(\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P} - \mathbf{P} \text{diag}(\mathbf{P}))\hat{\boldsymbol{\beta}}$, the correct check when every block is noised.

Remark 7.2 (When the expansion is valid). $\mathbf{P}\mathbf{E}$ is similar to the symmetric matrix $\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}$, so its spectral radius equals $\|\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}\|_{\text{op}} = \max_{\mathbf{u} \neq 0} |\mathbf{u}^\top \mathbf{E} \mathbf{u}| / (\mathbf{u}^\top \mathbf{G} \mathbf{u})$. The Neumann series for $(\mathbf{I} + \mathbf{P}\mathbf{E})^{-1}$ converges if and only if this is below one, i.e. $-\mathbf{G} \prec \mathbf{E} \prec \mathbf{G}$; on $\mathcal{E}_\eta$ it converges geometrically with ratio η . For the symmetric Gaussian mechanism $\|\mathbf{E}\|_{\text{op}}$ concentrates at or below the semicircle edge $2\sigma\sqrt{p}$ with sub-Gaussian tails, and $\|\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}\|_{\text{op}} \leq \|\mathbf{E}\|_{\text{op}}/\lambda_{\min}(\mathbf{G})$, so

$$\Pr(\mathcal{E}_\eta^c) \leq \exp(-p(\eta\rho - 1)^2) \quad (\eta\rho > 1).$$

Since $\lambda_{\min}(\mathbf{G}) = n\lambda_{\min}(\hat{\Sigma})$ grows linearly in n while σ is $O(1)$ in n , the condition holds with high probability once $n \gtrsim 2\sqrt{p}\sigma/(\eta\lambda_{\min}(\hat{\Sigma}))$. Below that threshold — small cohorts, strong privacy, or raw moments — the expansion is invalid and a ridge is required.

Remark 7.3 (The release gate). Algorithm 4 releases only if $\hat{\rho} = \lambda_{\min}(\tilde{\mathbf{G}} + \lambda\Psi)/(2\sigma\sqrt{p}) \geq 1$. On $\mathcal{E}_\eta$, $\tilde{\mathbf{G}} = \mathbf{G}^{1/2}(\mathbf{I} + \mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2})\mathbf{G}^{1/2} \succeq (1 - \eta)\mathbf{G}$, so $\hat{\rho} \geq (1 - \eta)\rho$ and the gate passes whenever $\rho \geq 1/(1 - \eta)$. Conditioning on the gate rather than on the symmetric event $\mathcal{E}_\eta$ therefore changes the conditional mean only by terms of order $\Pr(\mathcal{E}_\eta^c)$. The gate is post-processing of released values and costs no privacy.

7.3 Schur-complement reading

Let $\mathbf{S} = \mathbf{B} - \mathbf{C}^\top \mathbf{A}^{-1} \mathbf{C}$ be the Schur complement of $\mathbf{A}$ in $\mathbf{G}$. Block inversion gives $\Pi_O \mathbf{P} \Pi_O = \mathbf{S}^{-1}$, hence

$$\text{tr}(\Pi_O \mathbf{P}) = \text{tr}(\mathbf{S}^{-1}), \quad \text{tr}(\mathbf{P}) = \text{tr}(\mathbf{A}^{-1}) + \text{tr}(\mathbf{A}^{-1} \mathbf{C} \mathbf{S}^{-1} \mathbf{C}^\top \mathbf{A}^{-1}) + \text{tr}(\mathbf{S}^{-1}). \quad (15)$$

The two trace weights governing the leading bias are $\text{tr}(\mathbf{G}^{-1})$ for the O-coefficients and $\text{tr}(\mathbf{S}^{-1})$ for the R-coefficients.

Remark 7.4 (Bias and the distillation diagnostic). $\mathbf{S}$ is exactly the matrix of the distillation loss $\beta_O^\top \mathbf{S} \beta_O$ (Section 8.3): it measures the part of O's predictive contribution lying outside $\text{col}(\mathbf{X}_R^I)$. When O's features carry little signal beyond R's — the regime where distillation looks attractive — $\mathbf{S}$ is near-singular, $\text{tr}(\mathbf{S}^{-1})$ explodes, and the private estimator is worst-biased in R's own coefficients.

7.4 Ridge regularization with a fixed ridge

Let $\Psi \in \{\mathbf{I}, \Pi_O\}$, $\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi$, $\mathbf{P}_\lambda = \mathbf{G}_\lambda^{-1}$, $\hat{\beta}_\lambda = \mathbf{P}_\lambda \mathbf{c}$ (the non-private ridge target), and $\tilde{\beta}_\lambda = (\tilde{\mathbf{G}} + \lambda\Psi)^{-1} \tilde{\mathbf{c}}$. Let ρ_λ and $\mathcal{E}_{\eta,\lambda}$ be as in (13) with $\mathbf{G}$ replaced by $\mathbf{G}_\lambda$.

Theorem 7.2 (Ridge mechanism bias). *Suppose λ and Ψ are fixed independently of $(\mathbf{E}, \mathbf{f})$, and $\eta\rho_\lambda > 1$. Then*

$$\mathbb{E}[\tilde{\beta}_\lambda \mid \mathcal{E}_{\eta,\lambda}] - \hat{\beta}_\lambda = \sigma^2 M(\mathbf{P}_\lambda) \hat{\beta}_\lambda + O(\sigma^4) + O(e^{-p(\eta\rho_\lambda - 1)^2/2}),$$

with the moment-vector noise $\mathbf{f}$ again variance-only.

The proof is that of Theorem 7.1 under $\mathbf{G} \mapsto \mathbf{G}_\lambda$, which leaves $\mathbf{E}$, its covariance (12) and $\mathbf{f}$ untouched. The independence of λ from the noise is essential; see Section 7.5.

Proposition 7.1 (Total bias against OLS: shrinkage split). *If the estimand is $\hat{\beta} = \mathbf{G}^{-1} \mathbf{c}$, the total bias of the private ridge splits as*

$$\mathbb{E}[\tilde{\beta}_\lambda \mid \mathcal{E}_{\eta,\lambda}] - \hat{\beta} = \underbrace{\hat{\beta}_\lambda - \hat{\beta}}_{\text{deterministic}} + \underbrace{\mathbb{E}[\tilde{\beta}_\lambda \mid \mathcal{E}_{\eta,\lambda}] - \hat{\beta}_\lambda}_{\text{mechanism}}, \quad \hat{\beta}_\lambda - \hat{\beta} = -\lambda \mathbf{P}_\lambda \Psi \hat{\beta},$$

by the resolvent identity $(\mathbf{G} + \lambda\Psi)^{-1} - \mathbf{G}^{-1} = -\lambda \mathbf{P}_\lambda \Psi \mathbf{G}^{-1}$. The deterministic term grows in λ ; the mechanism term decays like σ^2/λ .

Both ridges control the bias: $\mathbf{P}_\lambda$ has eigenvalues $(g_j + \lambda)^{-1}$, and the ridged Schur complement satisfies $\mathbf{S}_\lambda \succeq \lambda \mathbf{I}$, so $\text{tr}(\Pi_O \mathbf{P}_\lambda) \leq d_O/\lambda$ and, for full ridge, $\text{tr}(\mathbf{P}_\lambda) \leq p/\lambda$. They differ in validity: for $\Psi = \mathbf{I}$, $\lambda_{\min}(\mathbf{G}_\lambda) = \lambda_{\min}(\mathbf{G}) + \lambda$ grows without bound; for $\Psi = \Pi_O$, $\lambda_{\min}(\mathbf{G} + \lambda \Pi_O) \rightarrow \lambda_{\min}(\mathbf{A})$ as $\lambda \rightarrow \infty$, so the O-block ridge cannot validate the expansion when $\lambda_{\min}(\mathbf{A}) \leq 2\eta^{-1}\sigma\sqrt{p}$, and near that ceiling the λ it needs grows without bound.

7.5 Choosing the ridge without looking at the noise

Earlier drafts chose (λ, Ψ) from the released $\tilde{\mathbf{G}}$ to reach a target $\hat{\rho}$. That choice is post-processing and costs no privacy, but it makes λ a function of $\mathbf{E}$ at first order, so Theorem 7.2 no longer applies: the fluctuation of λ contributes $O(\sigma^2)$ terms that the correction does not remove. In simulation, with a fixed λ the correction removed 87–98% of the mechanism bias, whereas under the original adaptive gate it removed essentially none: the residual after correction was 1.0–1.6 times the uncorrected bias across ridge-regime settings (and three to five times under a stylized rule that sets $\lambda_{\min}(\tilde{\mathbf{G}} + \lambda \mathbf{I})$ equal to the target). Near the O-block ceiling the adaptive rule also selected very large λ and shrank β_O to nearly zero.

We therefore fix the ridge from public inputs before decryption:

$$\Psi = \mathbf{I}, \quad \lambda = \max\{0, 2\rho^*\sigma\sqrt{p} - n\ell\}, \quad (16)$$

where $\rho^* > 1$ is the validity target and $\ell \geq 0$ is a committed public lower bound on $\lambda_{\min}(\mathbf{G}/n)$ for the standardized design (with $\ell = 0$, (16) is a certificate that always reaches $\rho_\lambda \geq \rho^*$). If ℓ is optimistic, the release gate of Remark 7.3 reports it. In simulation $\rho^* \approx 2.5$ made gate refusals negligible, and $\ell = 0$ left the ridge on at every n , with poor interval coverage (Section 9.3). The O-block ridge may be used with a λ likewise fixed in advance when a committed lower bound on $\lambda_{\min}(\mathbf{S})/n$ is available.

7.6 Zero-budget correction

Since σ , d_R , d_O and Π_O are public and $\tilde{\mathbf{G}}$ is already private, the bias is estimable by post-processing. With $\tilde{\mathbf{P}}_\lambda = (\tilde{\mathbf{G}} + \lambda \Psi)^{-1}$,

$$\hat{\beta}_{\text{bc}} = \tilde{\beta}_\lambda - \sigma^2 M(\tilde{\mathbf{P}}_\lambda) \tilde{\beta}_\lambda. \quad (17)$$

Substituting $\tilde{\mathbf{G}}$ for $\mathbf{G}$ inside a term already of order σ^2 injects only $O(\sigma^3)$ fluctuation and $O(\sigma^4)$ bias, so $\hat{\beta}_{\text{bc}}$ is conditionally unbiased for β_λ through $O(\sigma^4)$; by post-processing it carries the same guarantee. One must not additionally invert the deterministic shrinkage: de-shrinking multiplies coordinate j by $(g_j + \lambda)/g_j$ and re-amplifies precisely the directions the ridge protects. The honest deliverable is $\hat{\beta}_{\text{bc}}$ with the shrinkage $-\lambda \tilde{\mathbf{P}}_\lambda \Psi \tilde{\beta}_\lambda$ reported separately. In practice the bias removed is small; Section 9.3 shows that the correction acts mainly as a variance reduction around $\hat{\beta}_\lambda$.

7.7 Bridge to inference, magnitude and regime

The first-order expansion behind Theorem 7.1 is $\tilde{\beta} \approx \hat{\beta} + \mathbf{P}\mathbf{f} - \mathbf{P}\mathbf{E}\hat{\beta}$, with $\mathbf{P}\mathbf{f}$ and $\mathbf{P}\mathbf{E}\hat{\beta}$ uncorrelated. A variance decomposition $\text{Var}(\tilde{\beta}) \approx s^2\mathbf{P} + (\text{mechanism block})$ follows, with the residual scale estimated from $\widetilde{\text{RSS}} = \widetilde{\mathbf{y}^\top \mathbf{y}} - 2\tilde{\mathbf{c}}^\top \tilde{\beta} + \tilde{\beta}^\top \tilde{\mathbf{G}} \tilde{\beta}$ (floored at zero; it needs a small Jensen correction). Valid standard errors are developed in a companion note.

For a single release the mechanism standard deviation is $O(\sigma)$ while the bias is $O(\sigma^2)$, so the first-order inference error is undercoverage from ignoring the mechanism variance, not the biased point estimate. Both DP effects scale like $1/n^2$ against the sampling term's $1/n$, so they are asymptotically negligible but not in the small-to-moderate cohorts typical of research-data-centre use. In simulations with $d_R = d_O = 5$ uniform features, $B_R = B_O = \sqrt{5}$ and $B_y = 5$, at $\varepsilon = 1$ the validity target forced a full ridge unless $n \gtrsim 1.7 \times 10^4$, and a 10% relative error required $n \approx 3.6 \times 10^4$ (about 5.8×10^3 at $\varepsilon = 8$); committed-scale standardization is the largest lever on these numbers. Section 9.2 repeats this study with realistic feature scales and committed quantile bounds.

8 The Inference Gap: Prediction After Training

8.1 The fundamental problem

After Protocol 4.1 completes, R holds $\hat{\beta} = (\beta_R; \beta_O)$. For a new individual i , $\hat{y}_i = \mathbf{x}_{R,i}^\top \beta_R + \mathbf{x}_{O,i}^\top \beta_O$; R can evaluate the first term but has no access to $\mathbf{x}_{O,i}$. The model is not self-contained from R's perspective. We describe two approaches.

8.2 Approach 1: split inference via encrypted model weights

R retains β_R and sends $\text{Enc}(\beta_O)$ to O once. Per batch, after linkage on the test identifiers, O computes for each matched test record

$$\text{Enc}(p_{O,i}) = \bigoplus_{k=1}^{d_O} X_{O,ik} \cdot \text{Enc}(\beta_{O,k}), \quad (18)$$

and returns it; R decrypts and adds $\mathbf{x}_{R,i}^\top \beta_R$.

Sanitization is mandatory here. Without it, the reply's c_1 component is $\sum_k [\Delta' X_{O,ik}] c_1^{(k)}$, a known linear combination of the c_1 components of R's own ciphertexts $\text{Enc}(\beta_{O,k})$. R can solve this linear system over $\mathbb{Z}_q$ (N_C equations, d_O unknowns) and recover $\mathbf{x}_{O,i}$ *exactly*, whatever noise is added to the value. O must therefore apply a fresh encryption and SANITIZECKKS to every $\text{Enc}(p_{O,i})$, and charge any per-query noise to the ledger.

Leakage through the decrypted value. Even with sanitization, each decrypted scalar $p_{O,i} = \mathbf{x}_{O,i}^\top \boldsymbol{\beta}_O$ is a known-direction projection: one linear equation in the d_O unknown entries of $\mathbf{x}_{O,i}$, from which R can form $\hat{\mathbf{x}}_{O,i}^{(1)} = p_{O,i} \boldsymbol{\beta}_O / \|\boldsymbol{\beta}_O\|^2$.

Attack 1: noise averaging across repeated queries. If O adds independent per-response noise $\varepsilon_t \sim \mathcal{N}(0, \sigma_{\text{inf}}^2)$ and individual i appears in k batches, averaging gives variance σ_{inf}^2/k . Calibration must use k_{max} , the maximum number of appearances of any individual, and the ledger must enforce it; under zCDP the cost of k_{max} queries is k_{max} times the per-query cost.

Attack 2: multi-direction reconstruction across model updates. If the model is retrained, yielding $\boldsymbol{\beta}_O^{(1)}, \dots, \boldsymbol{\beta}_O^{(T)}$, an individual queried under $T \geq d_O$ versions with a full-rank coefficient matrix $\mathbf{B}_T = [\boldsymbol{\beta}_O^{(1)} \mid \dots \mid \boldsymbol{\beta}_O^{(T)}]^\top$ can be reconstructed exactly in the noiseless case; for $T < d_O$ the least-norm solution $\mathbf{B}_T^\top (\mathbf{B}_T \mathbf{B}_T^\top)^{-1} \mathbf{p}_{O,i}$ improves monotonically with T . DP noise on each $\boldsymbol{\beta}_O^{(t)}$ and on each query limits this only if the ledger counts every retraining and every query.

8.3 Approach 2: knowledge distillation to an R-only model

A qualitatively different solution eliminates test-time interaction: R trains a student $\hat{\boldsymbol{\beta}}_R^{\text{st}} \in \mathbb{R}^{d_R}$ using only its own features and predicts $\mathbf{x}_{R,i}^\top \hat{\boldsymbol{\beta}}_R^{\text{st}}$ with no further communication.

Student coefficient

The student regresses the joint model’s fitted values $\tilde{\mathbf{y}} = \mathbf{Z} \hat{\boldsymbol{\beta}}$ on $\mathbf{X}_R^I$. With $H_Z = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top$ and $\text{col}(\mathbf{X}_R^I) \subseteq \text{col}(\mathbf{Z})$, $\mathbf{X}_R^I{}^\top \tilde{\mathbf{y}} = \mathbf{X}_R^I{}^\top \mathbf{y}^I = \mathbf{c}_R$. The first block row of the normal equations, $\mathbf{A} \boldsymbol{\beta}_R + \mathbf{C} \boldsymbol{\beta}_O = \mathbf{c}_R$, then gives

$$\hat{\boldsymbol{\beta}}_R^{\text{st}} = \mathbf{A}^{-1} \mathbf{c}_R = \boldsymbol{\beta}_R + \mathbf{A}^{-1} \mathbf{C} \boldsymbol{\beta}_O. \quad (19)$$

In the private protocol R computes $\mathbf{A}^{-1} \tilde{\mathbf{c}}_R$ from its exact $\mathbf{A}$ and the released $\tilde{\mathbf{c}}_R$; this is post-processing and costs no additional budget.

Distillation loss and viability diagnostic

Since $\tilde{\mathbf{y}} - \mathbf{X}_R^I \hat{\boldsymbol{\beta}}_R^{\text{st}} = \mathbf{M}_R \mathbf{X}_O^I \boldsymbol{\beta}_O$, with $\mathbf{M}_R$ the annihilator of $\text{col}(\mathbf{X}_R^I)$,

$$\|\tilde{\mathbf{y}} - \mathbf{X}_R^I \hat{\boldsymbol{\beta}}_R^{\text{st}}\|^2 = \boldsymbol{\beta}_O^\top (\mathbf{B} - \mathbf{C}^\top \mathbf{A}^{-1} \mathbf{C}) \boldsymbol{\beta}_O = \boldsymbol{\beta}_O^\top \mathbf{S} \boldsymbol{\beta}_O, \quad (20)$$

the quadratic form of $\boldsymbol{\beta}_O$ against the Schur complement of $\mathbf{A}$. It is zero iff $\mathbf{X}_O^I \boldsymbol{\beta}_O \in \text{col}(\mathbf{X}_R^I)$. When it is small, Approach 2 is appropriate; when large, O’s features carry

signal that R’s cannot proxy and Approach 1 should be used. Because $\mathbf{y}^I$ is unavailable to R, (20) is the only computable diagnostic. As a secondary consideration, $\hat{\beta}_R^{\text{st}}$ encodes the cross-correlation structure $\mathbf{A}^{-1}\mathbf{C}\beta_O$ and should be treated as a release.

9 Empirical Evaluation

We evaluate four things: the cryptographic layer of the earlier implementation, end to end (Section 9.1); the accuracy of the private estimator against the true coefficients on a synthetic population (Section 9.2); whether the zero-budget correction matters when σ is large (Section 9.3); and utility on real data, against AdaSSP [17] and as a function of the intersection size (Sections 9.4–9.5). Section 9.6 explains where the protocol loses utility. The scripts that produce every number below are in the `experiments/` directory of the project repository.

The revised sanitization (Section 6) is not yet implemented. Sections 9.2–9.6 therefore simulate the release of Algorithm 3 in plaintext: a single Gaussian draw per released entry, symmetric, on every O -dependent entry of $(\mathbf{G}, \mathbf{c})$, with $\mathbf{A}$ exact and σ from the analytic Gaussian mechanism [19]. Section 9.1 shows that encryption adds no error beyond this. The flooding noise of SANITIZECKKS and the discretization of the noise are not modelled; both are small by design (Sections 5.4 and 6).

9.1 The cryptographic layer is exact

We ran the earlier implementation of the protocol end to end, over sockets, on six cohorts ($n = 200$ to 64,000 matched records) with a plaintext backend, and on one cohort ($n = 400$ matched records out of $n_O = 4,000$) with OpenFHE 1.5.1 (BFV, $N = 16,384$, for the linkage; CKKS with scale 2^{50} for the aggregation). Labeled PSI recovered exactly the true match set in every run, with no misaligned rows. With the DP noise switched off, the released estimate agreed with its plaintext target to 2.9×10^{-10} with the plaintext backend and 1.7×10^{-10} with OpenFHE. The OpenFHE run took about 13 s end to end. The 20-of-20 ciphertext distinguishing attack of Section 6 was also run with OpenFHE. All error in the private estimate therefore comes from the DP noise and the ridge it forces.

9.2 Coefficient accuracy against intersection size

Setup. The population has $d_R = 4$ and $d_O = 6$ features on realistic raw scales (a score, a binary indicator, lognormal income and assets, a count and an age), Toeplitz-correlated with coefficient 0.3. They are standardized with public population constants and clipped to committed 99.5% bounds ($B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$). The generating slopes give $R^2 \in \{0.3, 0.5, 0.9\}$. The replace-one sensitivity is $\Delta = 86.2$; with $\delta = 10^{-5}$ the analytic Gaussian gives $\sigma = 606, 322, 172, 93, 52$ for

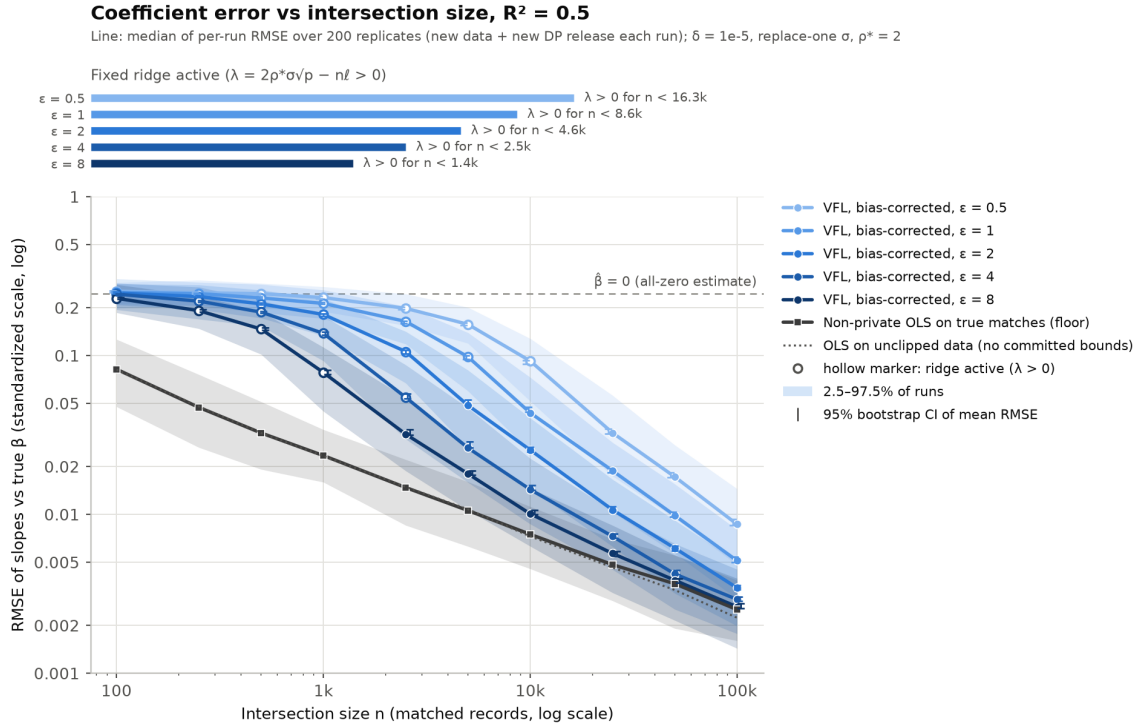


Figure 2: RMSE of the bias-corrected slopes against the true coefficients (standardized scale), against the intersection size n , at $R^2 = 0.5$. Lines: median over 200 runs, each with a new cohort and a new release. Bands: 2.5–97.5% of runs. Error bars: bootstrap 95% CI of the mean. Hollow markers: the ridge is active. Grey: non-private OLS on the true matches.

$\varepsilon = 0.5, 1, 2, 4, 8$. The ridge follows (16) with $\rho^* = 2$, $p = 11$ and the committed bound $\ell = 0.8 \lambda_{\min}(\mathbf{G}/n) = 0.494$ of the population. For each $n \in [10^2, 10^5]$ we draw 200 cohorts and an independent release per ε , and report the bias-corrected estimate (17).

Results. Figure 2 and Table 3 show three regimes. While the ridge is active ($n < 16.3k, 8.6k, 4.6k, 2.5k$ and $1.4k$ for $\varepsilon = 0.5, \dots, 8$), the estimate is shrunk towards zero. Once the ridge switches off, the DP error falls faster than the sampling error and merges into the OLS floor. The error reaches twice the floor at $n \approx 3.2k$ ($\varepsilon = 8$), $9.4k$ ($\varepsilon = 4$) and $34k$ ($\varepsilon = 2$); at $\varepsilon = 1$ the ratio is still 2.04 at $n = 10^5$ (about 105k by extrapolation). A stronger signal lowers the floor but not the DP error, so the ratio is largest at $R^2 = 0.9$, where the same crossings are at 13k, 30k and 66k.

Coefficients on different scales. Raw-scale errors cannot be compared across coefficients: on this population they span 5.6 orders of magnitude purely through units, and 87% of a pooled raw MSE comes from the two binary coefficients. We therefore

Table 3: Median RMSE of the standardized slopes against the true coefficients, $R^2 = 0.5$, with the ratio to the non-private OLS floor in brackets.

n	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	OLS floor
10^3	0.231 (9.9)	0.213 (9.1)	0.181 (7.7)	0.138 (5.9)	0.078 (3.3)	0.0234
10^4	0.092 (12.3)	0.044 (5.8)	0.025 (3.4)	0.014 (1.9)	0.010 (1.3)	0.0075
10^5	0.0087 (3.4)	0.0051 (2.0)	0.0034 (1.4)	0.0029 (1.2)	0.0026 (1.0)	0.0025

Efficiency ratio per coefficient: DP error in units of that coefficient's own OLS standard error, $R^2 = 0.5$

$SE_{OLS,j}$ = empirical sd of non-private OLS over the same replicates. 1 = as good as OLS. Black line = OLS's own RMSE/SE (above 1 only where clipping bias shows).

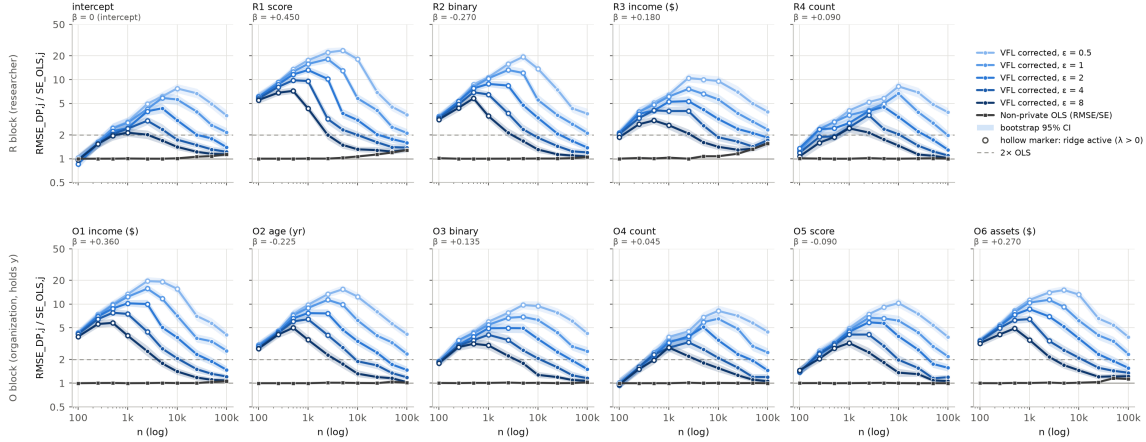


Figure 3: Efficiency ratio $RMSE_{DP,j}/SE_{OLS,j}$ per coefficient against n , $R^2 = 0.5$. Top row: R's block; bottom row: O's block.

report the *efficiency ratio* $RMSE_{DP,j}/SE_{OLS,j}$, which does not change when x_j or y is rescaled and has a direct reading: a ratio r means the release is about as accurate as OLS on n/r^2 records (Figure 3). Once $\lambda = 0$ the ratios agree to within about $\pm 15\%$ across coefficients, with O-block coefficients about 10% higher than R-block ones. While the ridge is on, the coefficients with the largest $|\beta_j|$ suffer most.

Inference. With $\lambda = 0$, Wald intervals built from the first-order variance (sampling plus mechanism) cover the true slopes 93.5–95.9% of the time for $n \leq 5 \times 10^4$, and the plug-in standard errors match the empirical spread within 0.99–1.03. With the ridge on, coverage collapses for the large coefficients. At large n coverage of the generating coefficients declines for OLS and the private estimate alike, because clipping at the committed bounds moves the target; against the clipped-population target the same intervals cover 93–96%. Coverage studies at large n should report both targets.

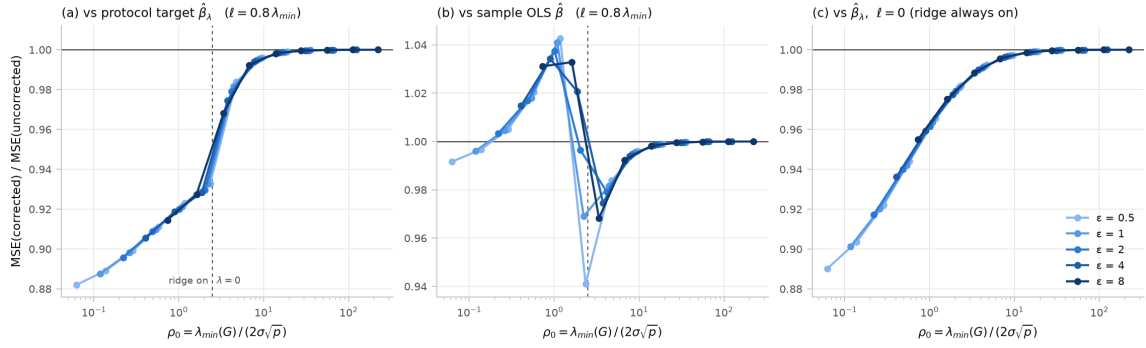


Figure 4: MSE of the bias-corrected estimate divided by that of the uncorrected one, against $\rho_0 = \lambda_{\min}(\mathbf{G})/(2\sigma\sqrt{p})$, one line per ε . (a) Against the protocol’s estimand $\hat{\beta}_\lambda$; (b) against the sample OLS estimate; (c) with $\ell = 0$, so that the ridge never switches off. Values below 1 favour the correction.

9.3 Does the bias correction matter when σ is large?

On the same population (Figure 4), the answer is: not as a bias correction. In a single release the mechanism bias of Theorem 7.2 is at most 0.75% of the MSE, and the largest coefficient bias is at most 0.14 of its standard deviation. When σ is large the ridge is on, and 72–90% of the error against the true coefficients is deterministic shrinkage (Proposition 7.1), which the correction leaves alone by design.

The correction is still worth applying. Because $M(\tilde{\mathbf{P}}_\lambda)$ is computed from the noisy Gram matrix, the correction also removes part of the noise: against the protocol’s estimand $\hat{\beta}_\lambda$ it lowers the MSE by 7–12% when the ridge is on and by under 0.5% once $\rho_0 = \lambda_{\min}(\mathbf{G})/(2\sigma\sqrt{p}) > 10$. The gain follows $1 - \text{MSE}_{\text{bc}}/\text{MSE} \approx 0.36/\rho_\lambda^2$ (constant 0.35–0.47 over 45 settings), and it never increased this MSE in any setting. Against the sample OLS estimate or the true coefficients, with a moderate ridge on, it costs 0.3–4.3%, because it removes a noise-driven inflation that had offset part of the shrinkage. The bias itself matters only if many releases of the same data are averaged: at least about 130, growing like $23\rho_0^2$. On the real datasets of Section 9.4 the correction changed test MSE by between –3.7% and +2.3%, helping at small ε and slightly hurting at large ε (Table 5).

Two choices in (16) matter more than the correction. First, the release gate (Remark 7.3) refused up to 1.6% of releases at $\rho^* = 2$; a refused release still spends budget, and $\rho^* \approx 2.5$ makes refusals negligible. Second, $\ell = 0$ keeps the ridge on at every n : at $n = 1.28 \times 10^5$ and $\varepsilon = 0.5$, λ was still 10% of $\lambda_{\min}(\mathbf{G})$, and at that n Wald coverage was 42–81% across ε . A committed $\ell > 0$ should be used whenever a defensible bound is available.

Table 4: Test MSE at $\varepsilon = 1$, as a ratio to non-private ridge regression on the same folds. Bold: lowest private method.

Dataset	n	d	Non-private test MSE	AdaSSP (published)	AdaSSP (matched)	VFL, ours	VFL, uncorrected
housing	506	13	0.0396	1.80	2.07	1.97	1.95
wine	1,599	11	0.0201	1.68	1.95	1.84	1.83
airfoil	1,503	5	0.0533	1.09	1.17	1.16	1.16
concrete	1,030	8	0.0444	1.48	1.74	1.69	1.68
bike	17,379	17	0.0279	1.03	1.07	1.06	1.06
elevators	16,599	18	0.00868	1.51	1.78	1.71	1.70
pol	15,000	26	0.135	1.04	1.08	1.06	1.06
kin40k	40,000	8	0.0632	1.00	1.00	1.00	1.00
protein	45,730	9	0.119	1.01	1.01	1.01	1.01

9.4 Benchmark against AdaSSP

AdaSSP [17] perturbs the same sufficient statistics and solves a ridge whose penalty adapts to a private estimate of $\lambda_{\min}(\mathbf{X}^\top \mathbf{X})$. It assumes one curator holding the joined table, which our setting rules out, so the comparison measures the utility cost of never pooling the data.

Setup. We use the 29 UCI datasets [18] of Wang’s published evaluation, with his preprocessing (z-scored features, rows scaled to unit norm, $y/\max|y|$), 10-fold cross-validation, test MSE, $\delta = 10^{-6}$ and 50 noise draws per fold. Our re-implementation of Wang’s procedures reproduces his published test MSE (median ratio 1.000 over 29 datasets and 10 values of ε ; range 0.75–1.46). R holds the first $\lfloor d/2 \rfloor$ features. The preprocessing gives $\|\mathbf{x}_R\|, \|\mathbf{x}_O\|, |y| \leq 1$, so $B_R = B_O = B_y = 1$ and $\Delta = \sqrt{10}$. No public lower bound on $\lambda_{\min}$ is available for these designs (several are rank-deficient), so $\ell = 0$. We compare four private methods:

- *AdaSSP (published)*: Wang’s algorithm and constants. It protects the whole record $(\mathbf{x}, y)$, a stronger guarantee than ours.
- *AdaSSP (matched)*: AdaSSP’s adaptive ridge under our adjacency and accountant, with $\mathbf{A}$ exact; $\varepsilon/3$ is spent on $\lambda_{\min}$. It is an idealized reference, since the private $\lambda_{\min}$ step needs $\mathbf{X}^\top \mathbf{X}$ in the clear.
- *VFL, ours*: the revised protocol with the correction (17); a draw that fails the release gate releases nothing ($\hat{\beta} = 0$).
- *VFL, uncorrected*: the same release and ridge without the correction.

Every method sees the same folds. Fold-to-fold variation is shared by all methods, so we compare methods by paired tests over folds (95% CI of the mean difference, t with 9 degrees of freedom).

Table 5: Paired comparisons over all 29 datasets: number of datasets where our protocol has significantly lower / higher test MSE (95% CI over folds). The remaining datasets show no significant difference.

Ours, compared with	$\varepsilon = 0.01$	$\varepsilon = 0.1$	$\varepsilon = 1$	$\varepsilon = 10$
vs AdaSSP (published)	21 / 0	14 / 8	6 / 19	1 / 21
vs AdaSSP (matched)	27 / 0	27 / 0	27 / 0	21 / 3
vs Uncorrected VFL	23 / 6	14 / 8	6 / 19	3 / 19

Table 6: Matched records needed to come within 10% of non-private test MSE: our protocol, with AdaSSP (published) in brackets. Log-linear interpolation of the mean curve; “>” means not reached at the sizes available.

Dataset	n available	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$
bike	15,641	>15k (15k)	11k (7.5k)	5.9k (3.8k)	3.1k (1.8k)
elevators	14,939	>14k (>14k)	>14k (>14k)	>14k (>14k)	>14k (14k)
pol	13,500	>13k (>13k)	10k (7.0k)	5.3k (3.5k)	2.8k (1.7k)
protein	41,157	5.4k (3.7k)	2.8k (1.7k)	1.4k (784)	619 (295)

Results. Tables 4–5 and Figure 5 give the comparison. At small ε our protocol is ahead of the published AdaSSP, which spends a third of its budget on $\lambda_{\min}$ (better on 21 datasets and worse on none at $\varepsilon = 0.01$). From $\varepsilon \approx 0.5$ AdaSSP is ahead: at $\varepsilon = 1$ our test MSE is a median 4% higher (range -6% to $+31\%$), and the gap shrinks with n (bike: 1.06 against 1.03; protein: 1.01 against 1.01). Under the same guarantee our protocol beats AdaSSP’s adaptive ridge on 27 of 29 datasets at $\varepsilon \leq 1$.

The published column should be read with one caveat. At $\varepsilon = 1$ AdaSSP adds noise of standard deviation 11.9 to each diagonal entry of $\mathbf{X}^\top \mathbf{X}$ and 8.4 to each off-diagonal entry, against our $\sigma = 13.4$, although its guarantee covers the whole record and it splits ε three ways. We have not reconciled this difference, which may reflect a looser adjacency in Wang’s calibration constant; until it is resolved, the published column may overstate AdaSSP’s advantage.

9.5 Intersection size on real data

Figure 6 repeats the experiment of Figure 2 on real data. We use the four datasets with at least 15,000 rows and a real linear signal: bike, elevators, pol and protein (kin40k is left out because its linear fit is no better than predicting zero). For each fold, the training folds are shuffled once and the first n rows form the matched set, so the subsets are nested across n ; the held-out fold is the test set. We report test MSE divided by non-private test MSE at the same n and fold, with 40 noise draws per fold and n . Table 6 gives the n at which each method comes within 10% of non-private.

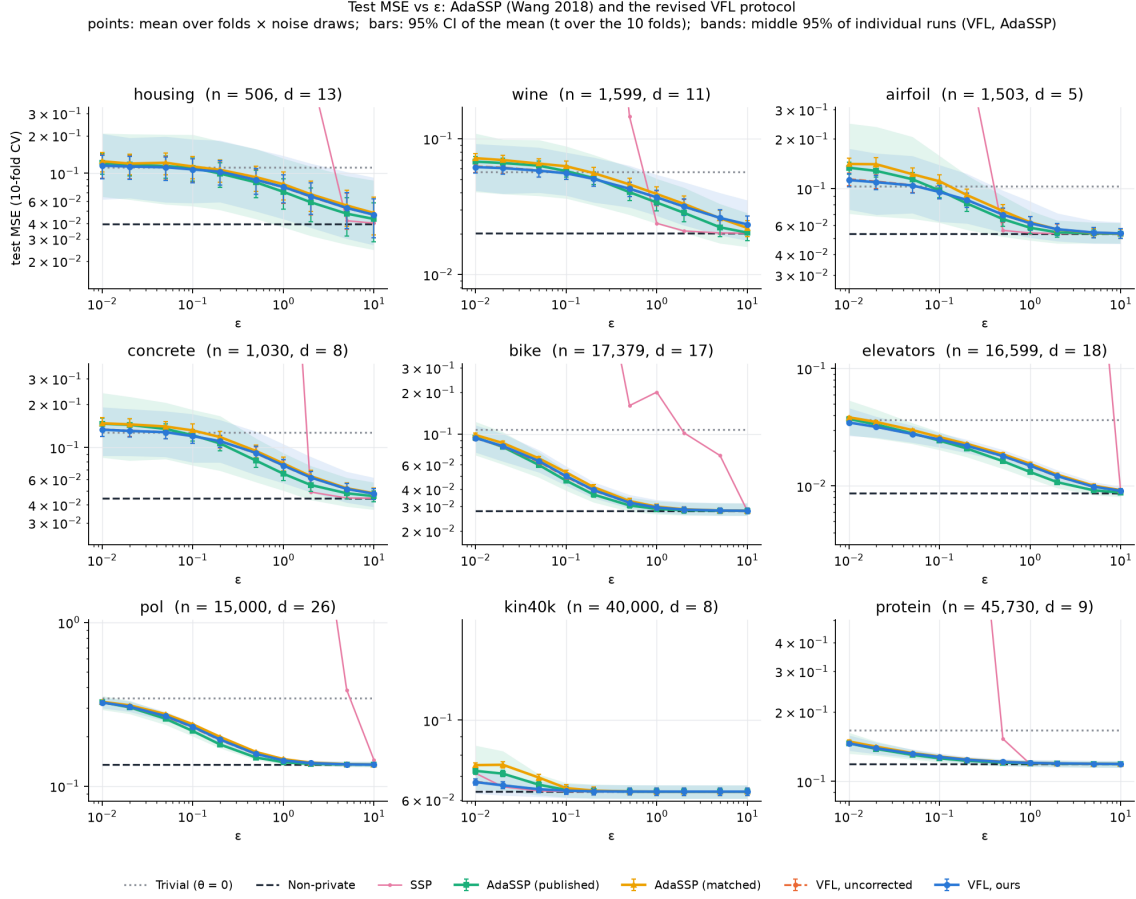


Figure 5: Test MSE against ϵ on nine of the datasets. Points: mean over 10 folds and 50 noise draws. Error bars: 95% CI of the mean (t over folds). Bands: middle 95% of individual runs.

Doubling ϵ roughly halves the records needed (bike: 11k, 5.9k, 3.1k for $\epsilon = 1, 2, 4$). Over this range of ϵ our protocol needs 1.4–2.1 times the records AdaSSP needs for the same relative cost. On bike, pol and protein, prediction comes within 10% of non-private at 1.4k–11k matched records for $\epsilon \in \{1, 2\}$. elevators never does at the sizes available: at its full 14.9k records our test MSE is still 1.7 times non-private at $\epsilon = 1$.

9.6 Where the utility goes

Prediction is not interpretation. Test MSE includes the irreducible noise, so it is forgiving: on real data the private fit is close to the non-private one at a few thousand matched records. The coefficients themselves, which are what a researcher asking “do my variables predict this outcome?” needs, take about ten times more

Privacy cost against intersection size on real data
mean over 10 folds x noise draws; bars: 95% CI over folds

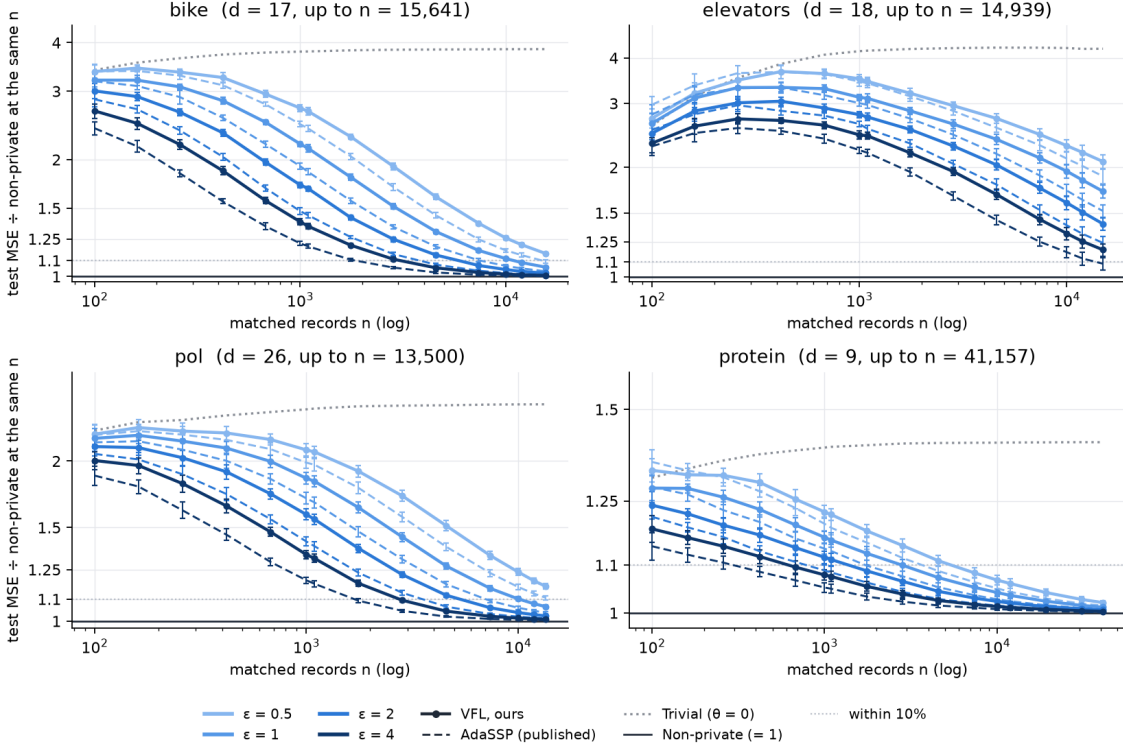


Figure 6: Test MSE divided by non-private test MSE at the same n , against the number of matched records. Solid: our protocol; dashed: AdaSSP (published); colour: ϵ . Error bars: 95% CI over folds.

data (Section 9.2: twice the OLS error at 34k records for $\epsilon = 2$ and about 105k for $\epsilon = 1$), and confidence intervals are valid only once the ridge is off.

The gap to AdaSSP is the noise level, not the ridge rule. Table 7 replaces the fixed ridge by an oracle one ($\ell = 0.8 \lambda_{\min}(\mathbf{G}/n)$, computed without privacy) and by no ridge. The oracle changes test MSE by at most 1.2% at $\epsilon = 1$; only at $\epsilon = 10$ on a well-conditioned design (wine) does it close most of the gap. Removing the ridge is far worse (27 times non-private on housing at $\epsilon = 1$). What separates the methods is σ . Our sensitivity bounds $\mathbf{x}_R$, $\mathbf{x}_O$ and y separately, while the preprocessing bounds the whole row, $\|(\mathbf{x}_R, \mathbf{x}_O)\| \leq 1$. A replace-one sensitivity under a joint row bound would reduce σ directly.

How well the data fill their bounds matters as much as n . The noise is sized for the committed bounds, so headroom the data do not use is lost signal. After

Table 7: Ridge ablation at full n (test MSE divided by non-private; 10 folds, 30 noise draws per fold).

Dataset	ε	AdaSSP (published)	VFL, $\ell = 0$	VFL, oracle ℓ	VFL, no ridge
housing	1	1.852	2.031	2.026	27
housing	10	1.084	1.180	1.163	2.55
wine	1	1.686	1.858	1.835	9.85
wine	10	1.014	1.162	1.058	1.03
elevators	1	1.514	1.722	1.722	2.37
elevators	10	1.013	1.051	1.051	1.28
bike	1	1.031	1.057	1.057	1.33
bike	10	1.001	1.002	1.002	1.28
pol	1	1.037	1.065	1.065	1.19
pol	10	1.001	1.003	1.003	1.01

Wang’s scaling by $\max |y|$, the typical $|y|$ is only 0.19 of its bound on elevators and 0.24 on wine, because a tail reaches 1.41 times the 99.5% quantile; on the other datasets it is 0.33–0.59. These are the two worst performers relative to their size. The protocol’s own preprocessing avoids the worst of this: bounds are committed from domain knowledge and enforced by clipping (Section 4), not set by the data maximum. Choosing bounds that typical records come close to, and units or transforms that make this possible (for example, log income), is part of using the protocol well.

Budget. Every release, including a refused one, is charged to $\mathcal{O}$ ’s ledger. Model search and exploratory re-runs therefore spend the budget quickly; the protocol suits a pre-specified model per research question.

10 Summary and Next Steps

We restricted the design to the case in which the organization holds the response and revised it so that its guarantees hold against an honest-but-curious researcher who holds the decryption keys: $\mathcal{O}$ ’s replies are sanitized, the noise matches the declared adjacency, calibration exchanges only committed constants, a ledger caps cumulative privacy loss, the linkage labels and metadata are data-independent, and the ridge is fixed from public inputs so that the bias theory applies. Empirically (Section 9), the private estimate is usable for prediction at a few thousand matched records for $\varepsilon \in \{1, 2\}$, coefficient-level conclusions need tens of thousands, the utility gap to centralized AdaSSP comes from the noise level rather than the ridge rule, and how well the data fill the committed bounds matters as much as n . Proposed next steps:

1. **Proof obligations.** Bound S_{err} (Obligation 6.1); complete the discrete-Gaussian and finite-precision accounting (Obligation 10.1); write out the conditional $O(\sigma^4)$

constants of Theorems 7.1–7.2.

2. **Implementation.** Implement the revised protocol in OpenFHE (fresh-encryption noise, flooding and modulus switching, per-party calibration, ledger, permutation and PSI padding, fixed ridge) and benchmark it as a function of n , d_R , d_O .
3. **Standard errors.** Develop the variance decomposition and coverage.
4. **Utility.** Derive the replace-one sensitivity under a joint row bound $\|(\mathbf{x}_R, \mathbf{x}_O)\| \leq B$, which would lower σ directly; use $\rho^* \approx 2.5$ and a committed $\ell > 0$ where defensible; quantify the gain from bounds that the data fill; and reconcile AdaSSP’s published noise constants with ours (Section 9.4).
5. **Extensions.** Logistic regression; the membership-private variant (Remark 3.1); malicious security (zero-knowledge proofs that $\mathbf{b} \in \{0, 1\}^{n_O}$ and $\|\mathbf{x}_{R,k}\| \leq B_R$).

Proof obligation 10.1 (Finite precision). The discrete Gaussian’s zCDP guarantee is for lattice-valued queries. The aggregates are real and are computed approximately under CKKS. The accounting must bound the effect of quantizing the statistic to $\gamma^{-1}\mathbb{Z}$ and of the CKKS approximation error on the sensitivity, and confirm that the sanitized decryption error is independent of the noise draw.

A Proof of Theorem 7.1

Throughout, condition on the sample so that $\mathbf{G}$, $\mathbf{c}$ and $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$ are fixed, and write $\mathbf{P} = \mathbf{G}^{-1}$ and $\mathbf{H} = \mathbf{P}\mathbf{E}$.

Step 1: Neumann expansion on $\mathcal{E}_\eta$. On $\mathcal{E}_\eta$ the spectral radius of $\mathbf{H}$ is at most $\eta < 1$ (Remark 7.2), so

$$\tilde{\boldsymbol{\beta}} = (\mathbf{G} + \mathbf{E})^{-1}(\mathbf{c} + \mathbf{f}) = (\mathbf{I} + \mathbf{H})^{-1}\mathbf{P}(\mathbf{c} + \mathbf{f}) = (\mathbf{I} - \mathbf{H} + \mathbf{H}^2 - \mathbf{H}^3 + \mathbf{H}^4(\mathbf{I} + \mathbf{H})^{-1})(\hat{\boldsymbol{\beta}} + \mathbf{P}\mathbf{f}),$$

and $\|\mathbf{H}^4(\mathbf{I} + \mathbf{H})^{-1}\|$ is bounded by a constant times $\|\mathbf{E}\|_{\text{op}}^4/(1 - \eta)$ in the $\mathbf{G}$ -norm.

Step 2: odd terms and the moment noise vanish. $\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$, $\mathbf{E}$ is symmetric about zero, and $\mathbf{f}$ is independent of $\mathbf{E}$ and symmetric about zero. Hence $\mathbb{E}[\mathbf{H}^m \mid \mathcal{E}_\eta] = 0$ for odd m , and every term containing $\mathbf{P}\mathbf{f}$ has zero conditional expectation. Therefore

$$\mathbb{E}[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta] = \hat{\boldsymbol{\beta}} + \mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta] \hat{\boldsymbol{\beta}} + O(\sigma^4),$$

the remainder being bounded by $\mathbb{E}[\|\mathbf{E}\|^4]/\Pr(\mathcal{E}_\eta)$ up to constants.

Step 3: removing the conditioning. $\mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta] - \mathbb{E}[\mathbf{H}^2] = (\mathbb{E}[\mathbf{H}^2 \mathbf{1}_{\mathcal{E}_\eta}] - \mathbb{E}[\mathbf{H}^2] \Pr(\mathcal{E}_\eta))/\Pr(\mathcal{E}_\eta)$, whose norm is at most $2(\mathbb{E}\|\mathbf{H}\|^4)^{1/2} \Pr(\mathcal{E}_\eta^c)^{1/2}/\Pr(\mathcal{E}_\eta) = O(\sigma^2 e^{-p(\eta\rho-1)^2/2})$ by Cauchy–Schwarz and Remark 7.2.

Step 4: the second-moment contraction. Write $\mathbf{p} = (P_{11}, \dots, P_{pp})^\top$ for the diagonal of $\mathbf{P}$ and $\mathbf{X} = (\chi_{ij})$.

Lemma A.1. *For any fixed symmetric $\mathbf{P}$, with $\circ$ the Hadamard product, $\mathbb{E}[\mathbf{EPE}] = \sigma^2(\mathbf{X} \circ \mathbf{P} + \text{diag}(\mathbf{Xp}) - \Pi_O \text{diag}(\mathbf{p}))$.*

Proof. Entrywise, $\mathbb{E}[(\mathbf{EPE})_{ab}] = \sum_{i,j} P_{ij} \mathbb{E}[E_{ai} E_{jb}]$, and (12) gives $\mathbb{E}[E_{ai} E_{jb}] = \sigma^2 \chi_{ai} (\delta_{aj} \delta_{ib} + \delta_{ab} \delta_{ij} - \delta_{ai} \delta_{aj} \delta_{ab})$. Summing against the three terms gives $\sigma^2 \chi_{ab} P_{ab}$, $\sigma^2 \delta_{ab} \sum_i \chi_{ai} P_{ii}$, and $-\sigma^2 \delta_{ab} \chi_{aa} P_{aa}$. With $\chi_{aa} = \mathbf{1}[a \in O]$ these are the (a, b) entries of $\mathbf{X} \circ \mathbf{P}$, $\text{diag}(\mathbf{Xp})$ and $-\Pi_O \text{diag}(\mathbf{p})$. $\square$

Writing $\chi_{ij} = 1 - (\Pi_R)_{ii}(\Pi_R)_{jj}$ turns the first two terms into projector form: $\mathbf{X} \circ \mathbf{P} = \mathbf{P} - \Pi_R \mathbf{P} \Pi_R$ and $\text{diag}(\mathbf{Xp}) = \text{tr}(\mathbf{P})\mathbf{I} - \text{tr}(\Pi_R \mathbf{P})\Pi_R$. Left-multiplying by $\mathbf{P}$ and using $\Pi_O \text{diag}(\mathbf{p}) = \text{diag}(\mathbf{P})\Pi_O$,

$$\mathbb{E}[\mathbf{PEPE}] = \sigma^2 [\mathbf{P}^2 - \mathbf{P} \Pi_R \mathbf{P} \Pi_R + \text{tr}(\mathbf{P})\mathbf{P} - \text{tr}(\Pi_R \mathbf{P})\mathbf{P} \Pi_R - \mathbf{P} \text{diag}(\mathbf{P})\Pi_O].$$

Step 5: regrouping. Substituting $\Pi_O = \mathbf{I} - \Pi_R$, the first four terms collapse to $(\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P})\Pi_O + (\mathbf{P}\Pi_O \mathbf{P} + \text{tr}(\Pi_O \mathbf{P})\mathbf{P})\Pi_R$, which gives $\mathbb{E}[\mathbf{H}^2] = \sigma^2 M(\mathbf{P})$. Combining with Steps 2 and 3 proves the theorem. $\square$

Degenerate check. With $\Pi_O = \mathbf{I}$ and $\Pi_R = \mathbf{0}$ the formula gives $\sigma^2(\mathbf{P}^2 + \text{tr}(\mathbf{P})\mathbf{P} - \mathbf{P} \text{diag}(\mathbf{P}))$; for $p = 1$ and $\mathbf{A} = a$ this is $\sigma^2 a^{-2} \hat{\beta}$, matching a direct expansion of $c/(a + e)$ with $\text{Var}(e) = \sigma^2$.

Validity of the plug-in correction. The map $\mathbf{P} \mapsto M(\mathbf{P})$ is smooth at $\mathbf{G}$; replacing $\mathbf{G}$ by $\tilde{\mathbf{G}}$ perturbs M by $O(\|\mathbf{E}\|_{\text{op}}) = O(\sigma)$. Since M is multiplied by σ^2 in (17), the induced error in $\hat{\beta}_{bc}$ is $O(\sigma^3)$ in the fluctuation and $O(\sigma^4)$ in conditional expectation.

Extension to ridge (Theorem 7.2). Replacing $\mathbf{G}$ by $\mathbf{G}_\lambda = \mathbf{G} + \lambda \Psi$ with (λ, Ψ) independent of $(\mathbf{E}, \mathbf{f})$ leaves $\mathbf{E}$, its covariance and $\mathbf{f}$ untouched, and Steps 1–5 hold verbatim with $\mathbf{P} \mapsto \mathbf{P}_\lambda$ and $\mathcal{E}_\eta \mapsto \mathcal{E}_{\eta, \lambda}$. If λ depended on $\mathbf{E}$, Step 2 would fail: $\mathbf{P}_\lambda$ would no longer be fixed under the conditional expectation.

B Labeled PSI with Record Alignment

The protocol requires $\mathbf{R}$ to learn I and the alignment π , while $\mathbf{O}$ learns only public sizes. We use labeled PSI in the style of Chen, Laine and Rindal [7, 8]: $\mathbf{R}$ encrypts its (hashed) identifiers, and $\mathbf{O}$ evaluates, on those ciphertexts, plaintext polynomials whose roots are its own identifiers and whose labels are $\mathbf{O}$'s row positions. The construction runs under BFV, whose exact arithmetic modulo t is needed for the zero test and

label recovery; CKKS approximation would not survive it. Algorithm 2 gives the pseudocode.

Pre-processing at O. O applies a uniformly random permutation $\tau \leftarrow \text{Sym}(n_O)$ to its rows; all subsequent indices, including the labels and the order used in Phase 2, refer to this ordering (Remark B.2). Identifiers are mapped to the field by a keyed hash $v = H_K(ID) \bmod (t - 1)$, which excludes the padding root $t - 1$. Each item is simple-hashed into all h candidate bins; each bin is split into α partitions of at most D items, where D and α are *public* functions of $(n_O, h, N_{\text{bins}}, d_{\text{cap}})$ chosen so that overflow has probability at most δ_{psi} . Every partition’s membership polynomial is padded with the root $t - 1$ to degree exactly D , and its label polynomial interpolates (value, position) pairs.

Query at R. R cuckoo-hashes [9] its hashed identifiers into N_{bins} bins with the same h hash functions, batching when n_R exceeds the table capacity (a load of at most $0.85N_{\text{bins}}$, or a stash, keeps insertion failures negligible), and sends BFV encryptions of the slot-wise powers $\mathbf{w}^{2^i}$, $i < W$, with $W = \lceil \log_2 D \rceil + 1$.

Evaluation and reply at O. O forms $\text{Enc}(\mathbf{w}^d)$, $d \leq D$, as products of window ciphertexts (depth $\lceil \log_2 W \rceil$), evaluates the membership and label polynomials of each partition as plaintext-coefficient linear combinations, masks with $\mathbf{r}_1, \mathbf{r}_2$ uniform in $(\mathbb{Z}_t^*)^{N_{\text{bins}}}$ from a CSPRNG, and returns

$$\text{SANITIZEBFV}(\mathbf{r}_1 \odot P(\mathbf{w})), \quad \text{SANITIZEBFV}(\mathbf{r}_2 \odot P(\mathbf{w}) + L(\mathbf{w})).$$

The decrypted values satisfy: the first is 0 in bin β iff R’s item there is in O’s set, and the second is then that item’s τ -position; otherwise both are uniform.

Recovery at R. For each occupied bin and partition, R sets $\pi(k)$ to the decrypted label when the flag is zero and the label is a valid position.

Remark B.1 (Collision probability). A false match requires two distinct identifiers to share a field value inside a common bin. The expected number is about $n_R n_O h / (N_{\text{bins}} t)$, so $t \geq n_R n_O h / (N_{\text{bins}} \delta_{\text{psi}})$ (a prime with $t \equiv 1 \bmod 2N_B$ for batching) keeps it below δ_{psi} . The padding root must be $t - 1$ for the t actually used; a root fixed from a different modulus falls inside the hash range and adds a false-match rate of about n_R/t .

Remark B.2 (Pre-shuffle necessity). Without τ , the positions $\pi(k)$ recovered by R would reflect O’s original database ordering (e.g. by region or registration date), leaking attributes outside the DP guarantee. The pre-shuffle makes positions semantically inert. It must be applied by O inside the protocol, not assumed of the input files.

Remark B.3 (Why not an encrypted-coefficient Horner evaluation). An earlier construction had R encrypt the coefficients of $\prod_k (x - a_k)$ and O evaluate it by

Horner’s rule. Under BFV, each Horner step multiplies the ciphertext noise by a scalar of size up to t , so n_R sequential steps are infeasible; the sum-of-powers form $\sum_k \text{Enc}(c_k) x^k$ grows noise only additively but still costs $O(n_R)$ operations per point. The labeled construction above has depth $O(\log D)$ with $D \leq d_{\text{cap}}$ and reveals no data-dependent metadata once (D, α) are public.

Remark B.4 (Linkage privacy). $\mathcal{O}$ sees only BFV ciphertexts under $\mathcal{R}$ ’s key and the number of query batches, so it learns neither I nor n — less than Definition 3.2 allows.

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